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Murray et al.

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(54) **STOVETOP FIRE SUPPRESSION SYSTEM
AND METHOD**

USPC 169/65, 26
See application file for complete search history.

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(74) *Attorney, Agent, or Firm* — Foley & Lardner LLP

(65) **Prior Publication Data**

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(57) **ABSTRACT**

Related U.S. Application Data

(60) Provisional application No. 63/062,920, filed on Aug.
7, 2020, provisional application No. 62/992,716, filed
on Mar. 20, 2020.

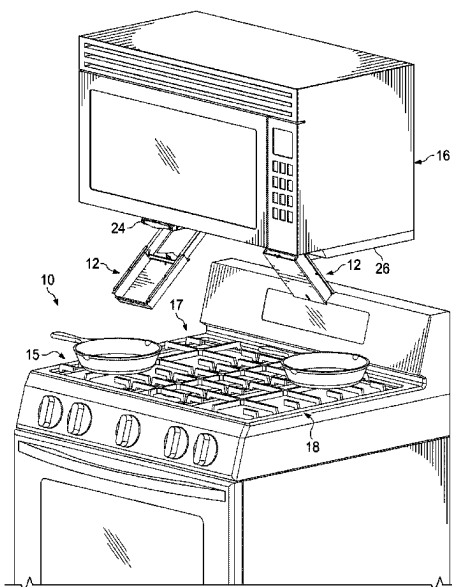
A stovetop fire suppression system for selectively releasing
a fire suppressant powder onto a stovetop first or second
burner. The stovetop fire suppression system includes a base
member and a chute member secured to the base member at
opposed first and second ends of the base member. The chute
member is movable between a storage position and a dis-
pensing position. The suppression system further includes a
first actuating member disposed at the base section first end
and a second actuating member disposed at the base section
second end. In response to actuation of the first actuating
member, the chute member is movable the dispensing posi-
tion to release the fire suppressant powder onto the first
burner, and in response to actuation of the second actuating
member, the chute member moves to the dispensing position
to release the fire suppressant powder onto the second
burner.

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A62C 99/00 (2010.01)

(52) **U.S. Cl.**
CPC **A62C 3/006** (2013.01); **A62C 99/0045**
(2013.01)

(58) **Field of Classification Search**
CPC A62C 3/006; A62C 35/02; A62C 35/04;
A62C 99/0045

20 Claims, 24 Drawing Sheets



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2019/0255371	A1 *	8/2019	Scarola	A62C 37/08
2021/0290995	A1 *	9/2021	Murray	A62C 99/0045

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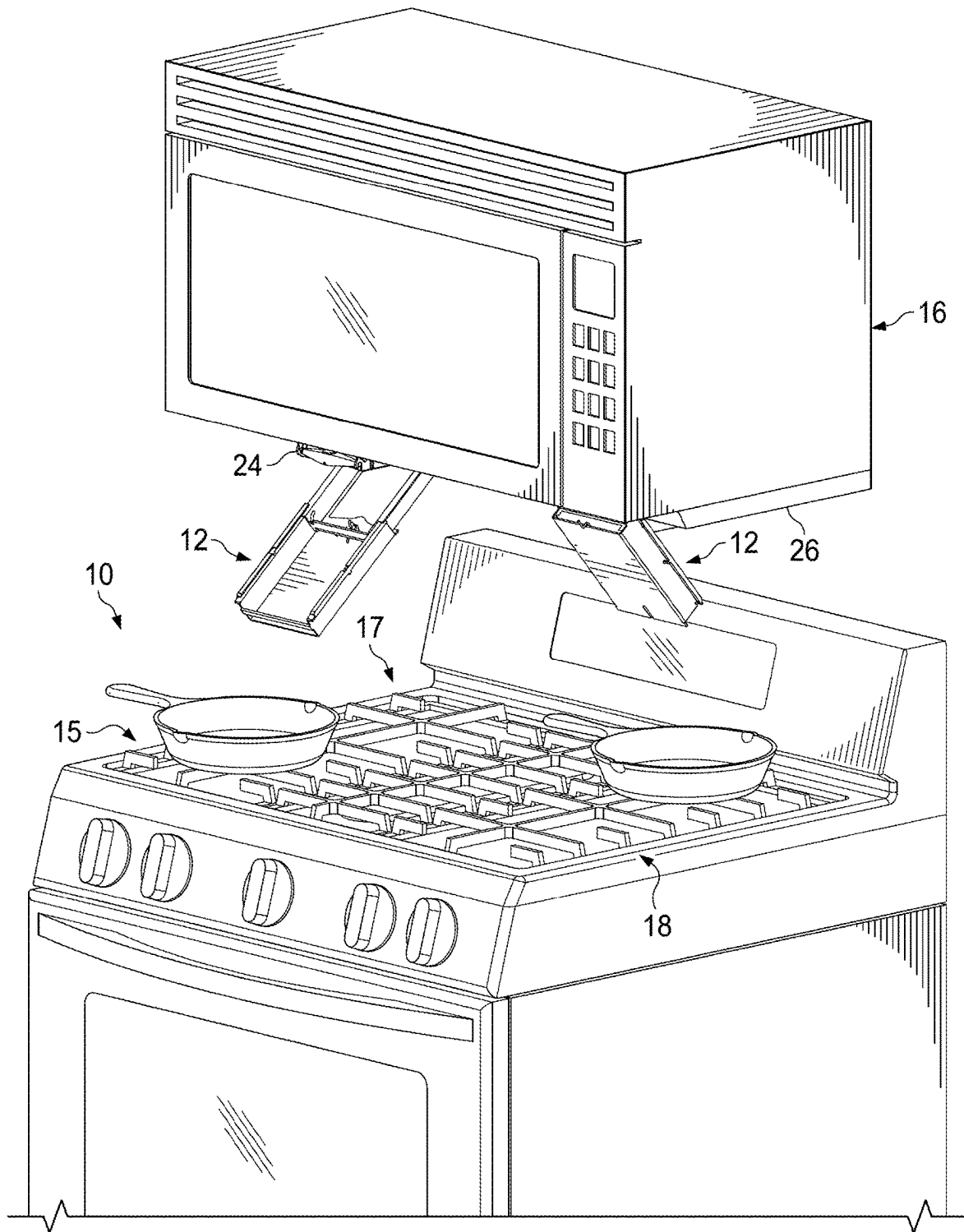


FIG. 1

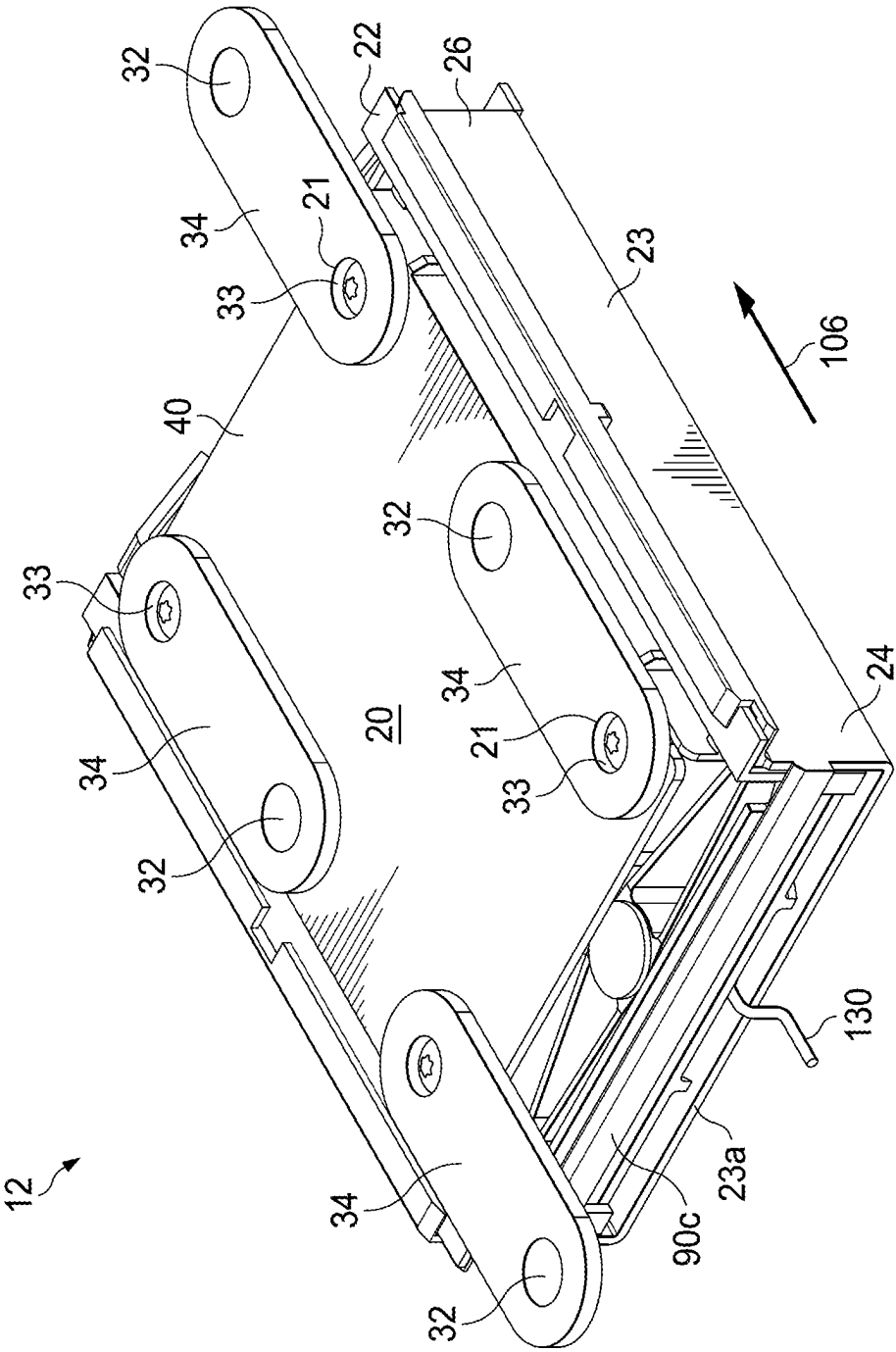


FIG. 2

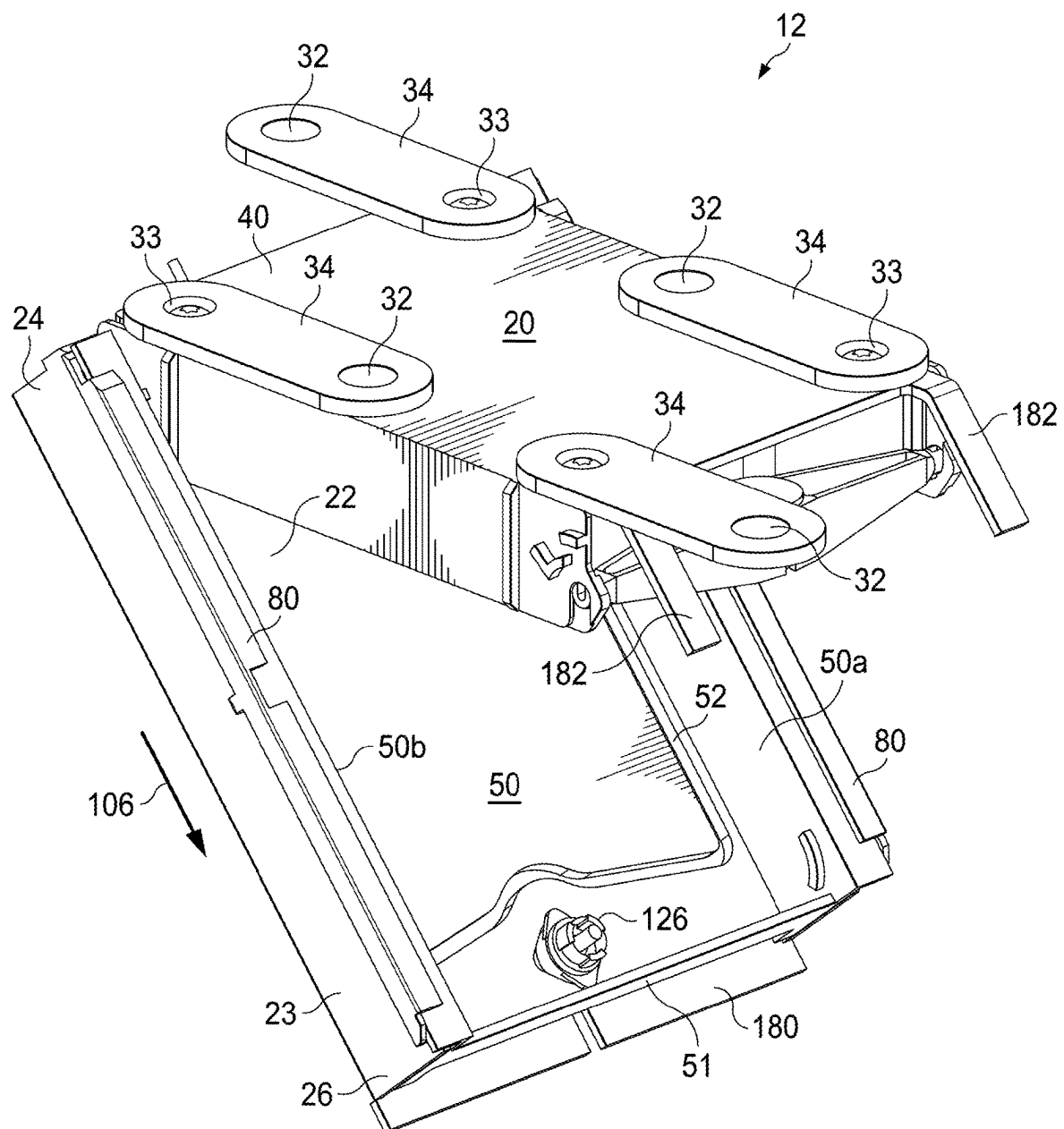


FIG. 3

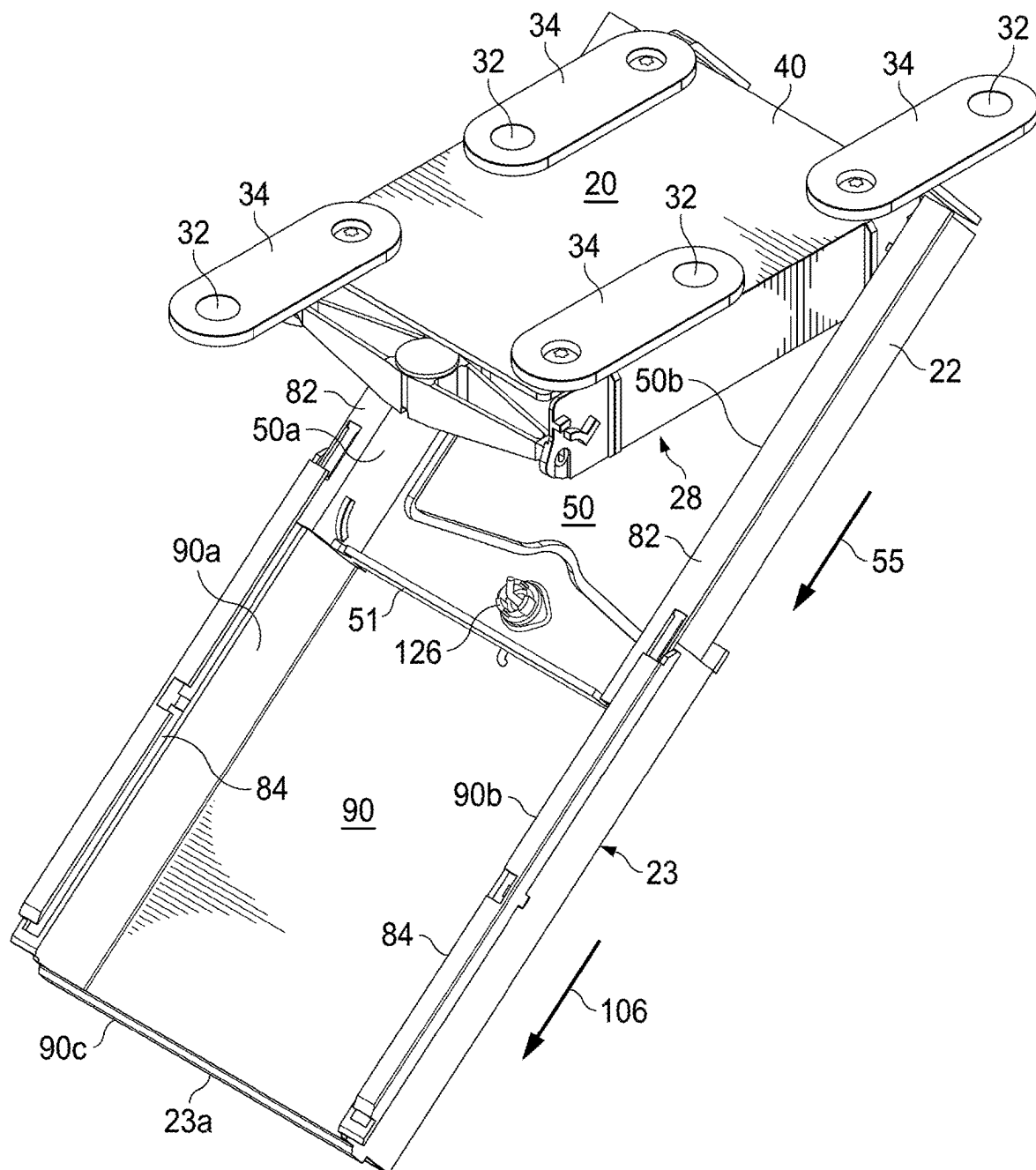


FIG. 4

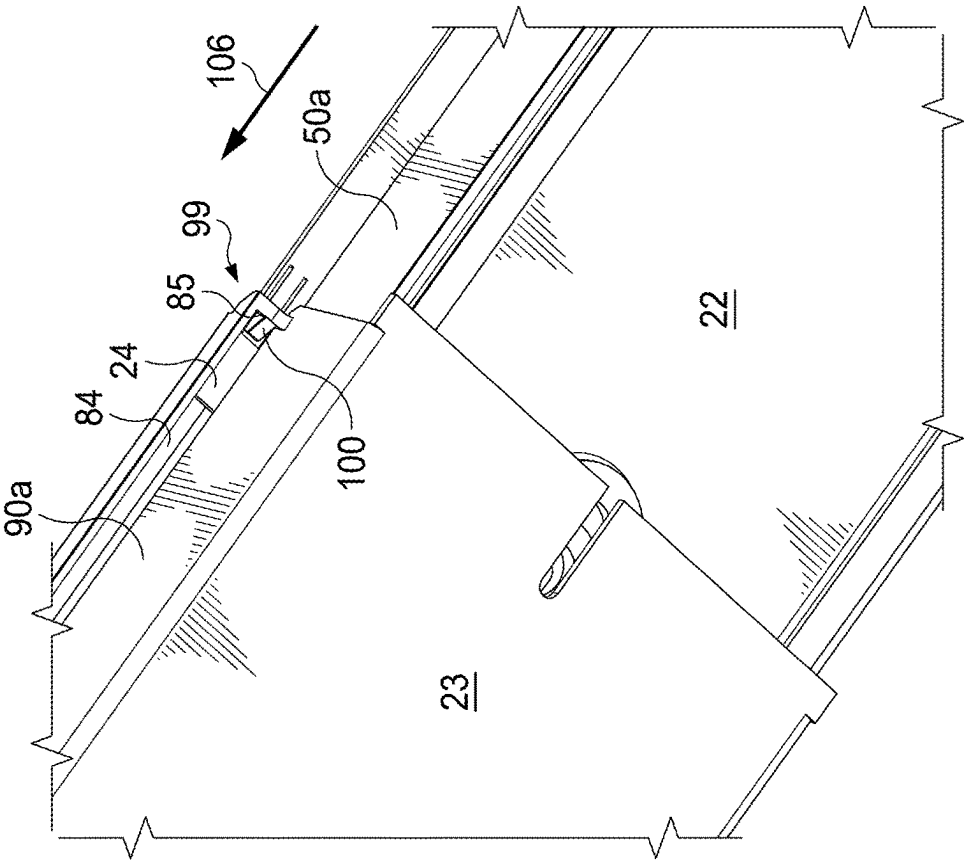


FIG. 4b

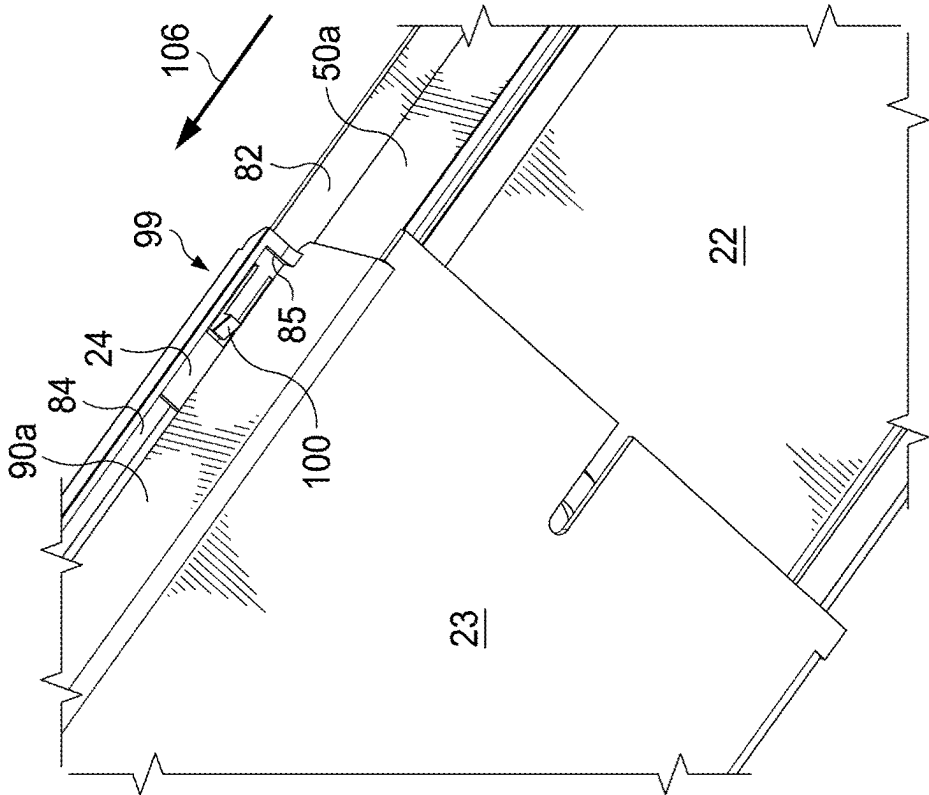


FIG. 4a

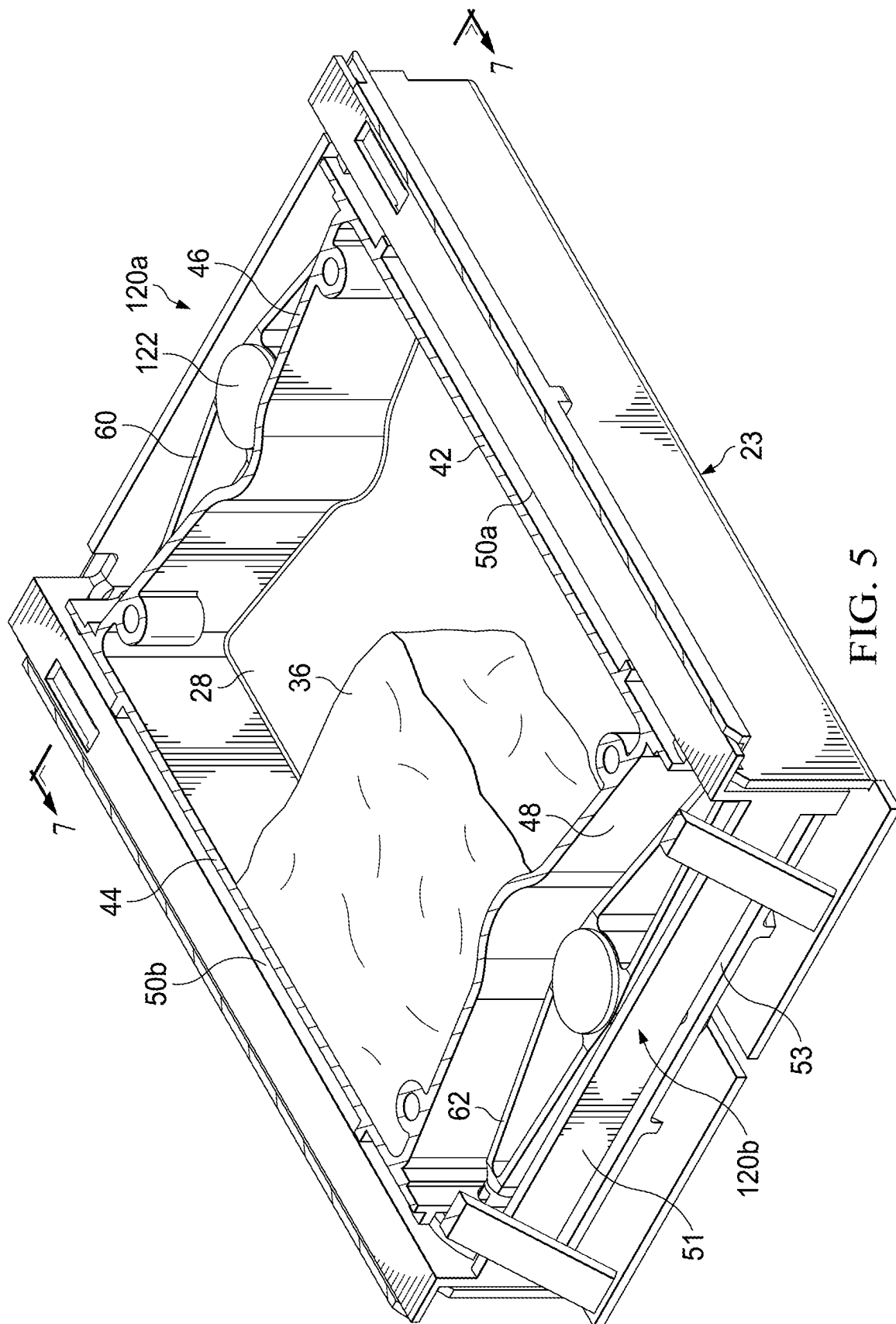


FIG. 5

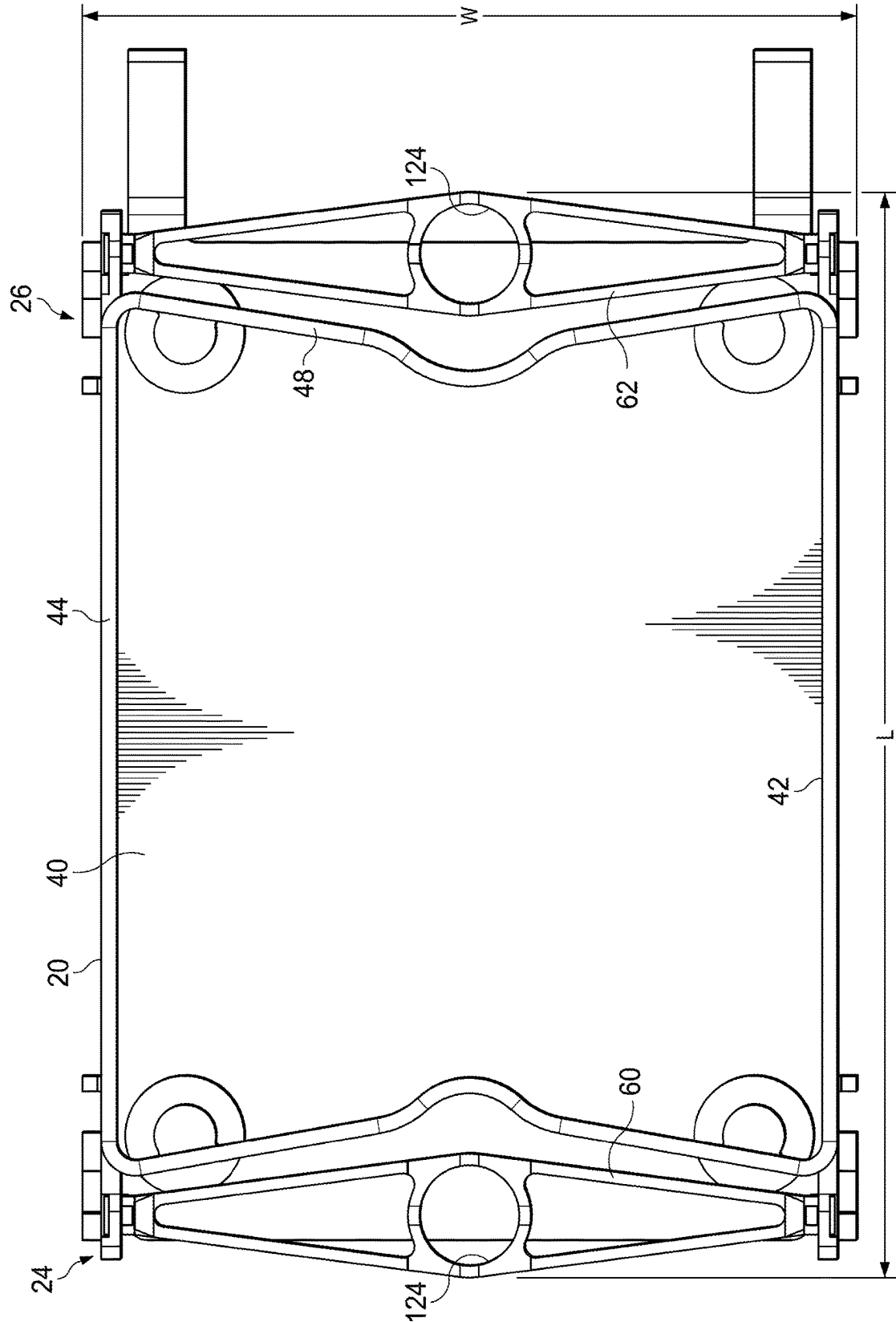
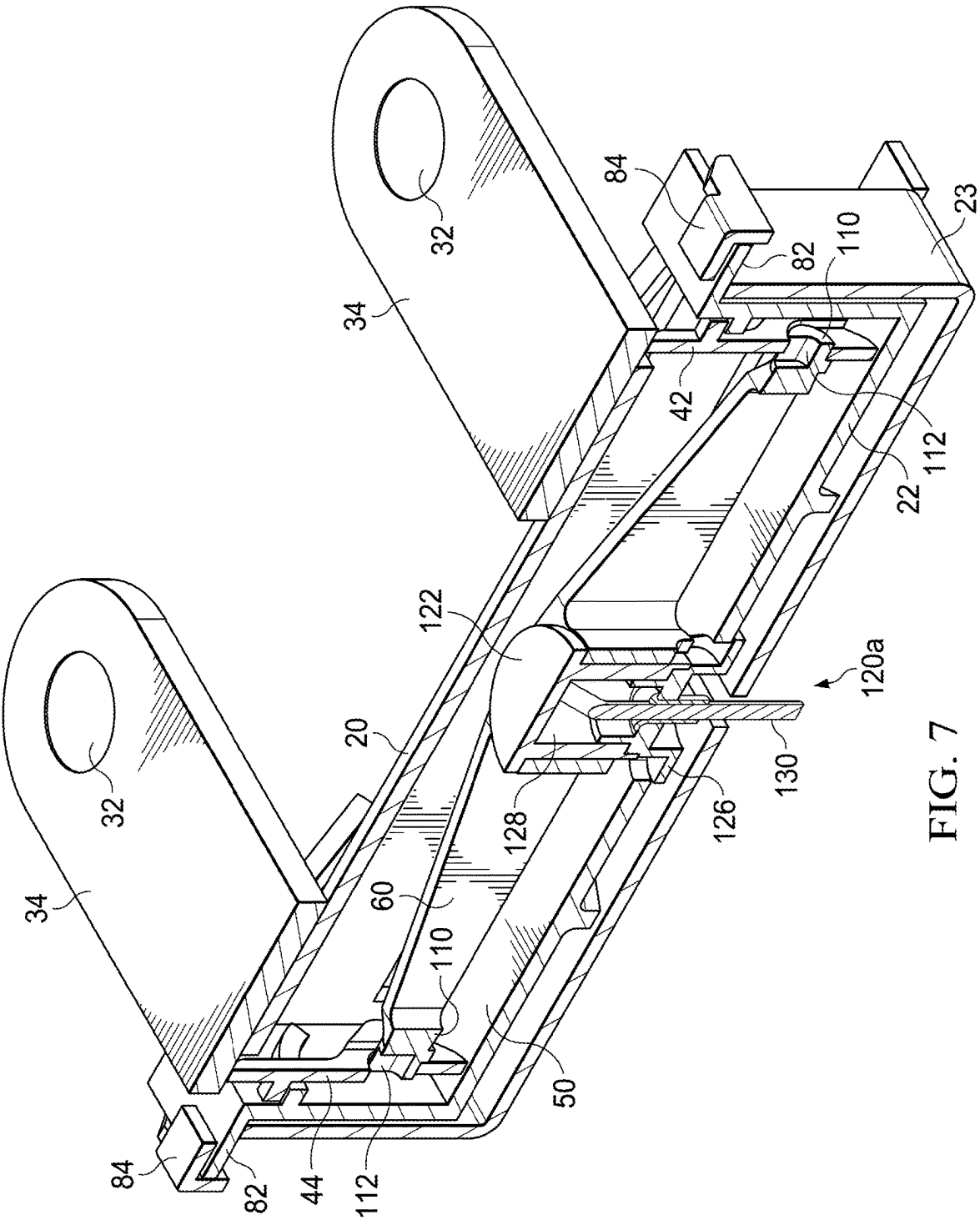


FIG. 6



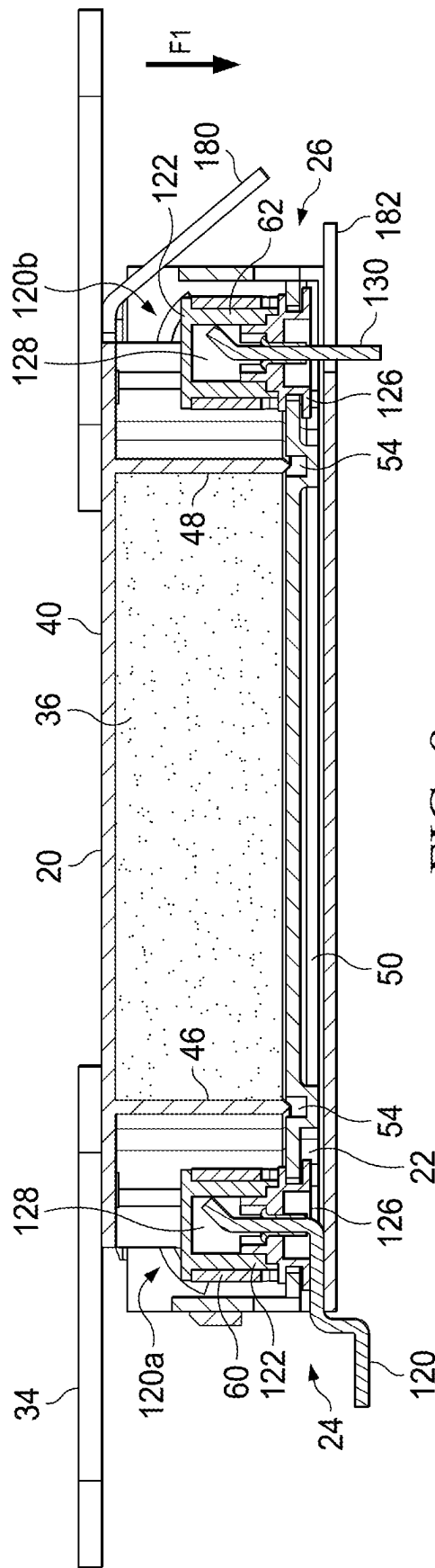
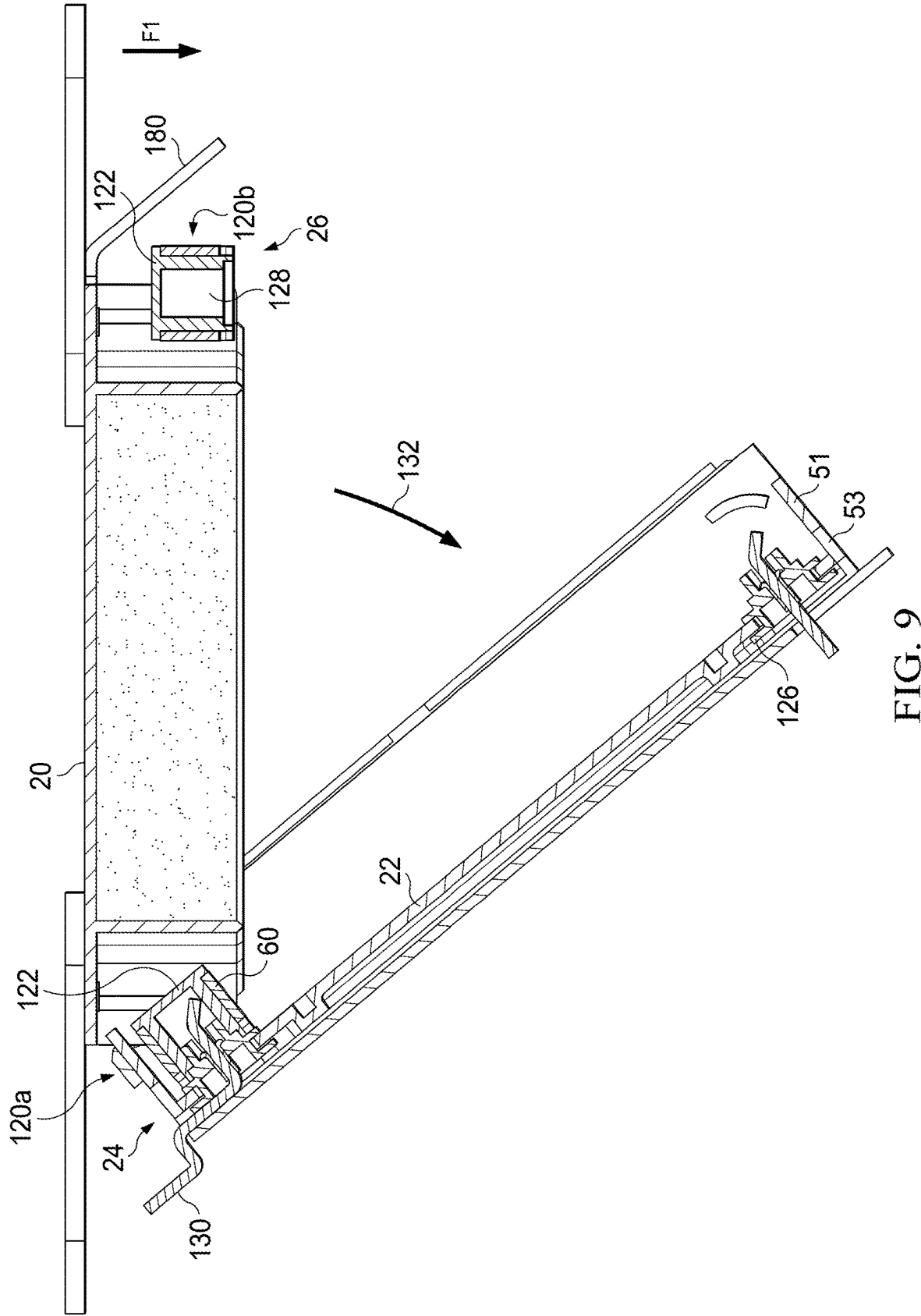
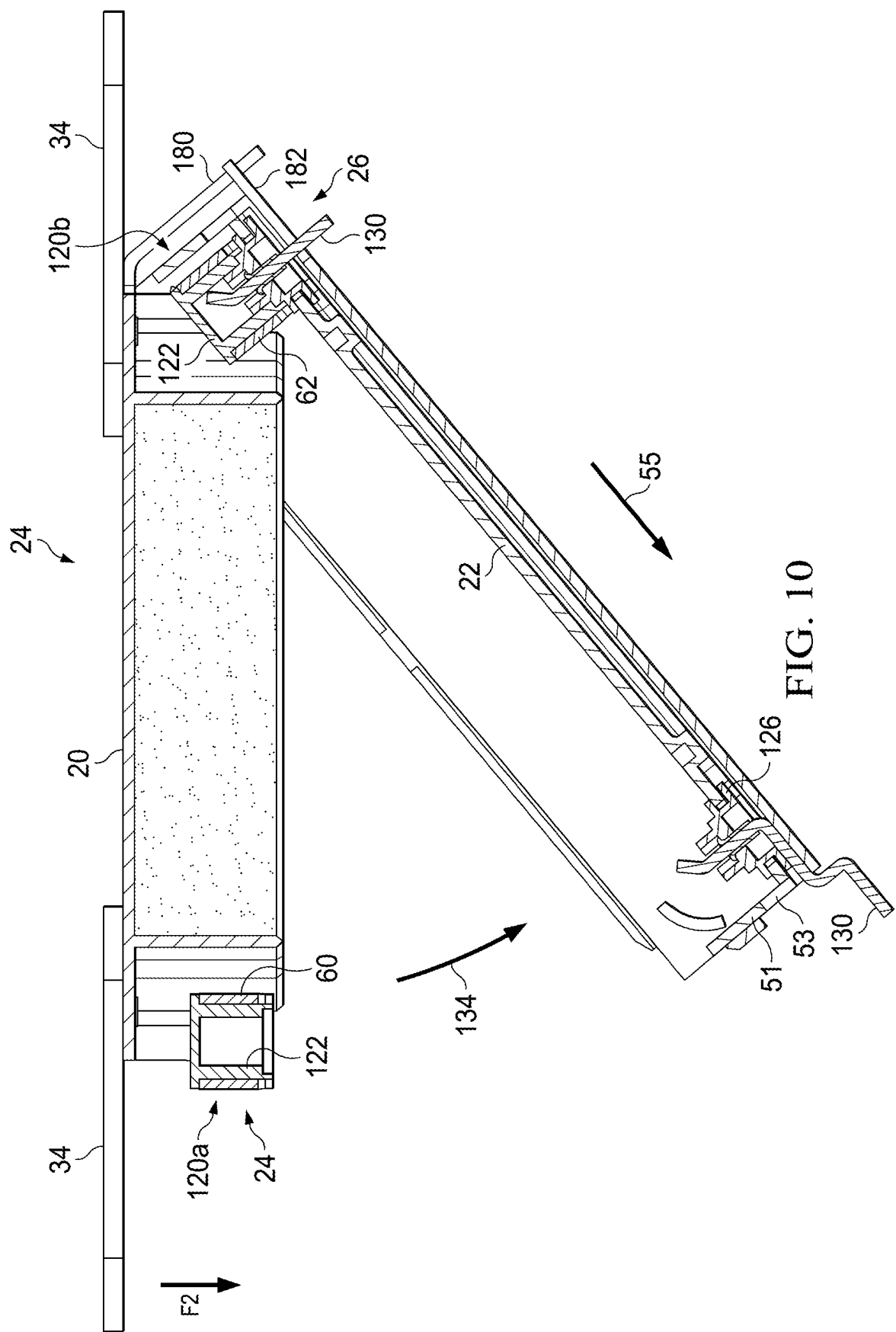


FIG. 8





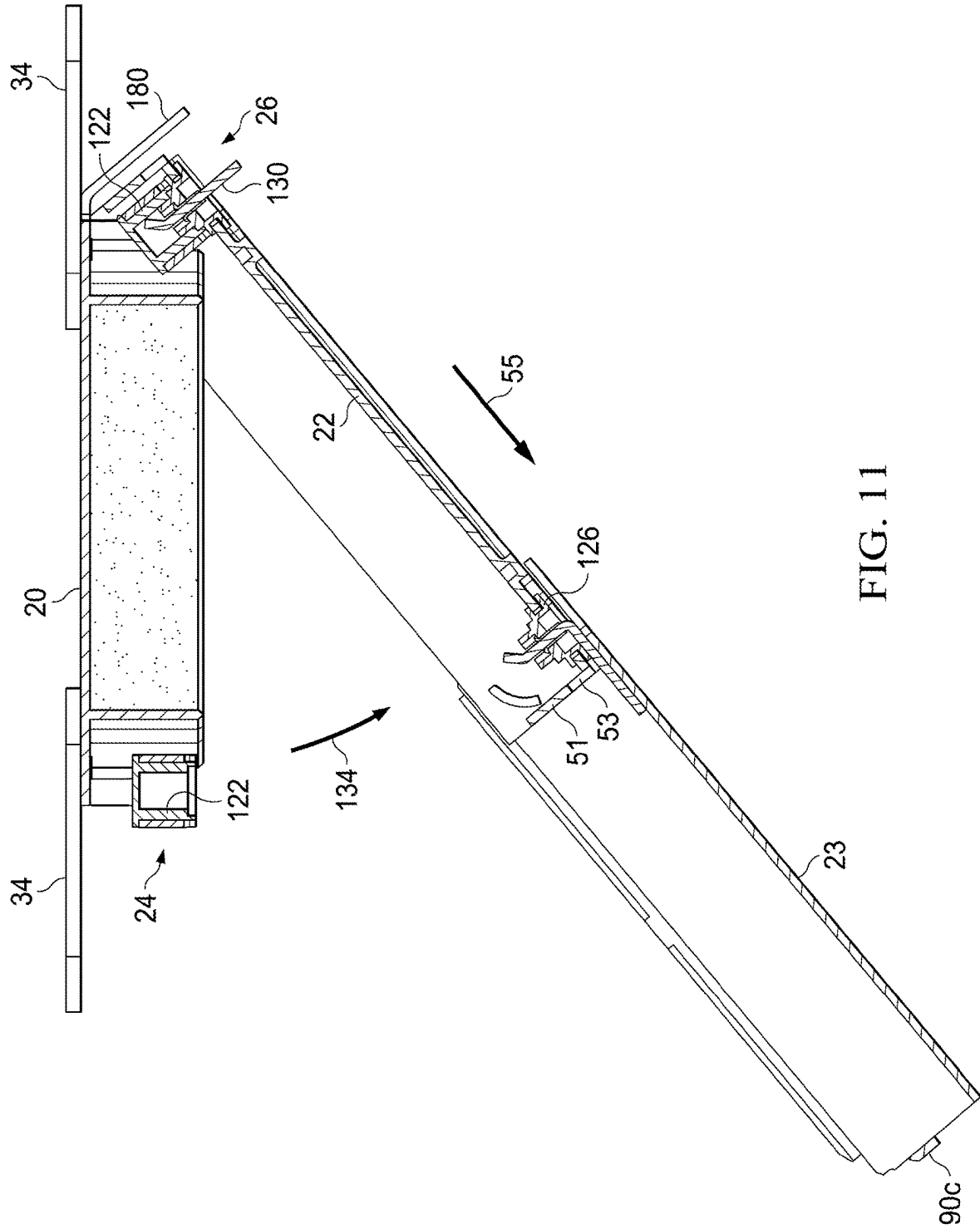


FIG. 11

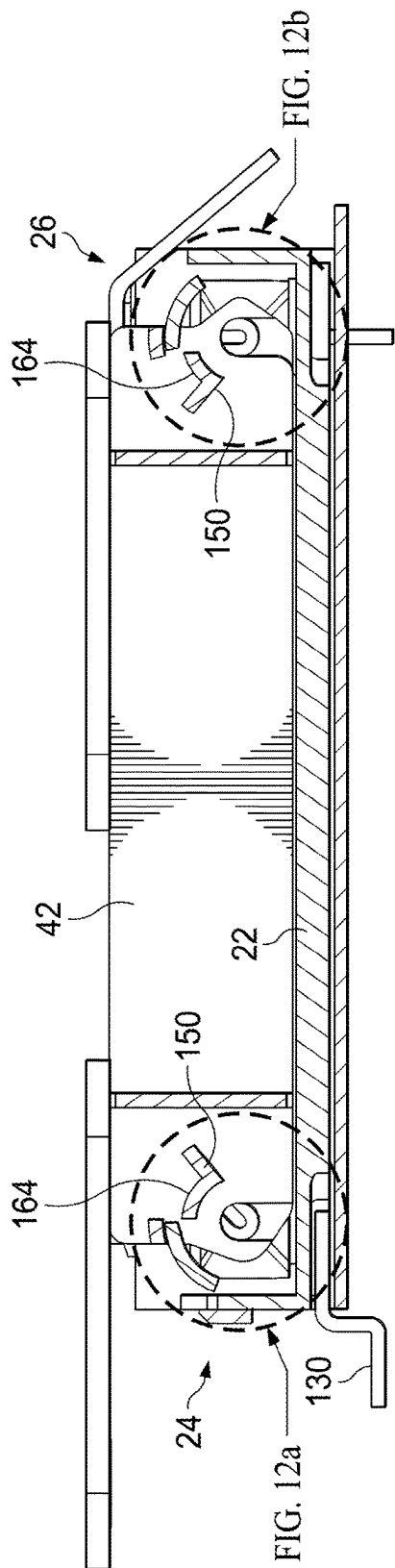


FIG. 12

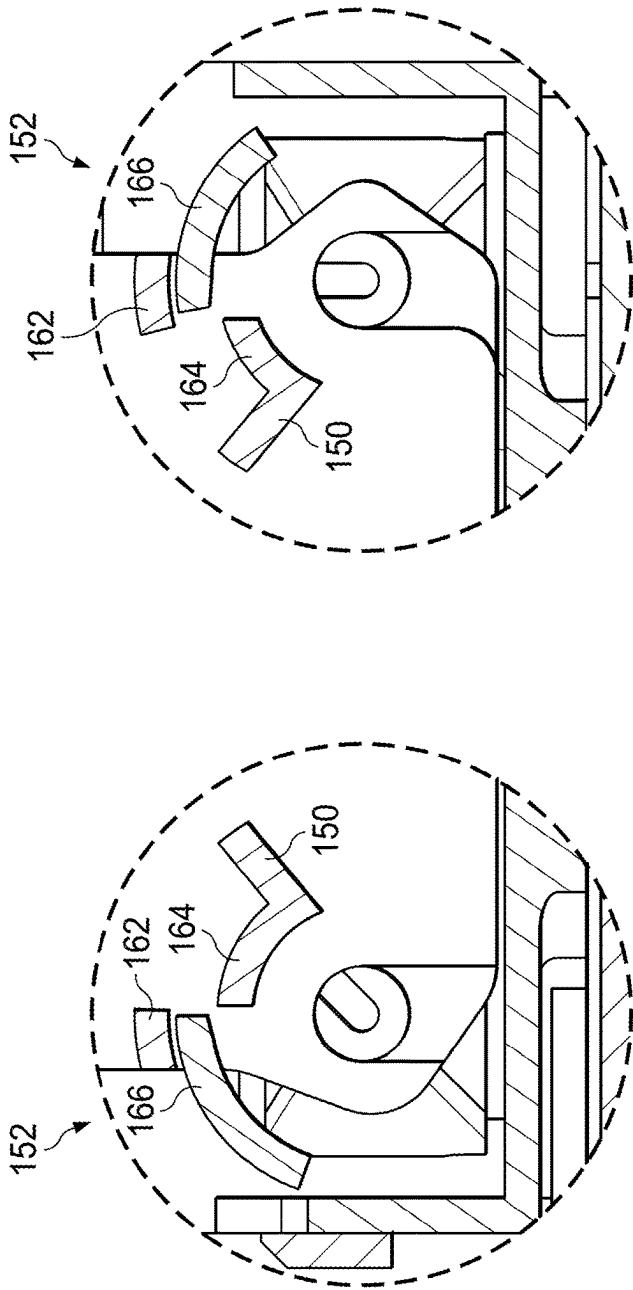
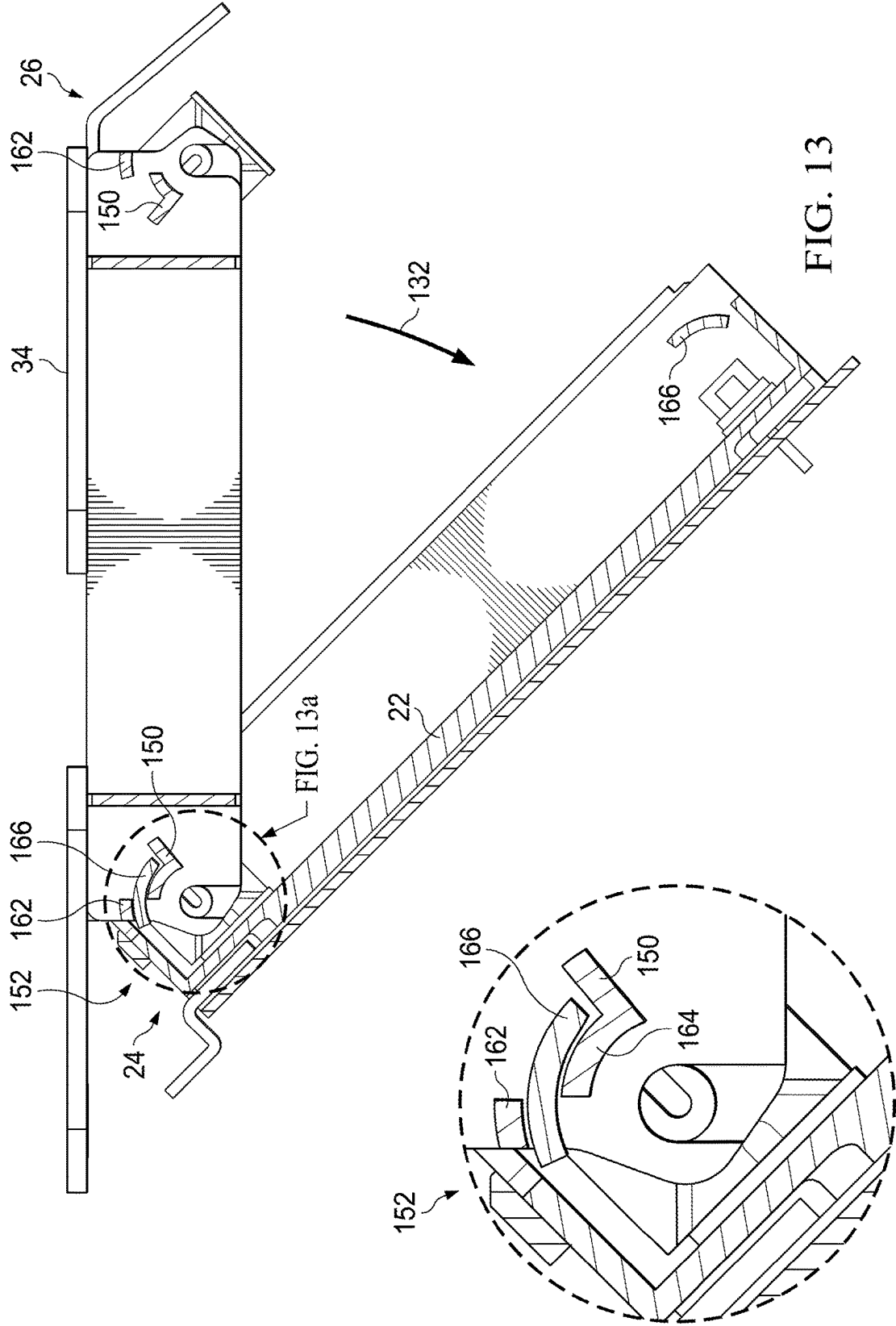
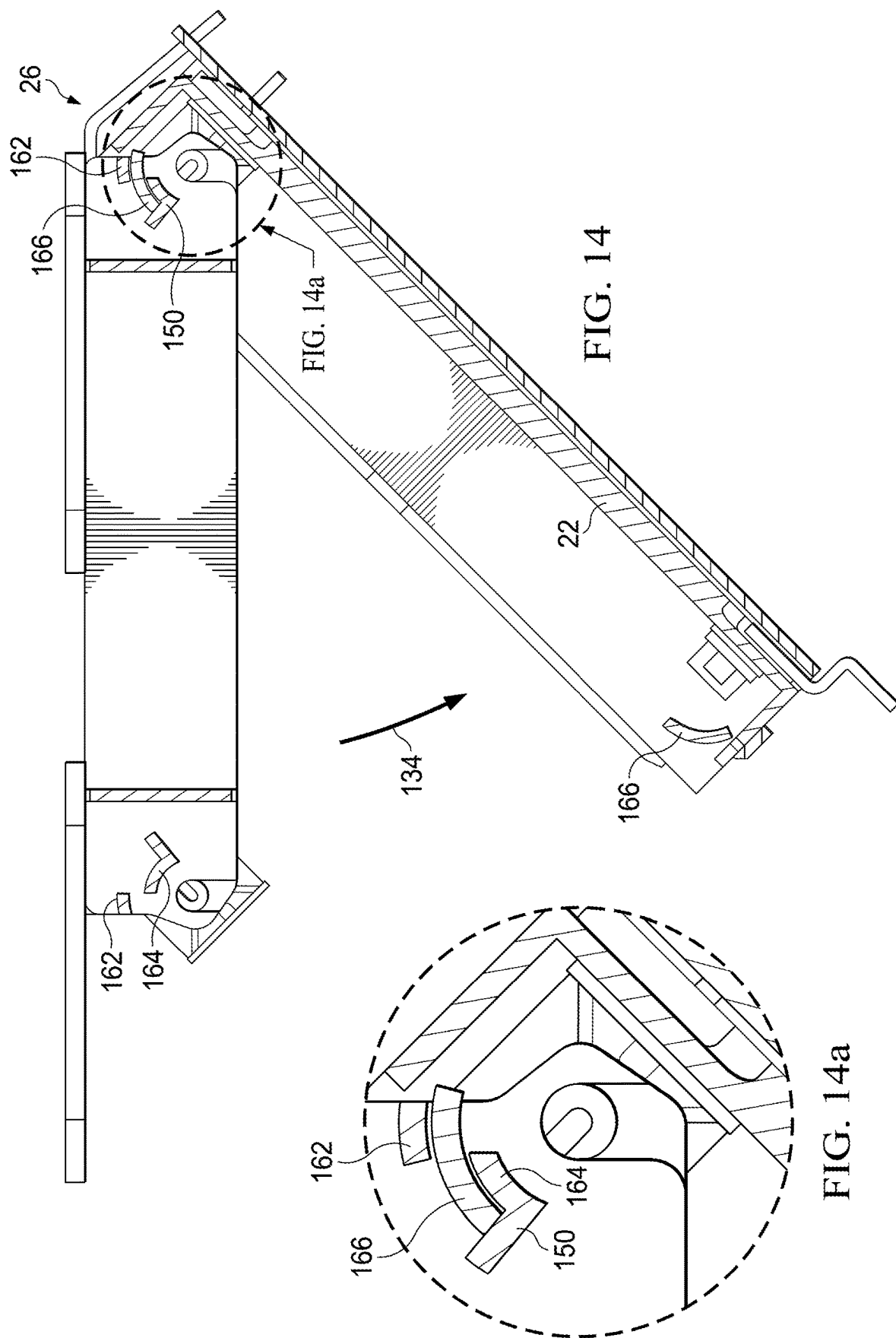


FIG. 12a

FIG. 12b





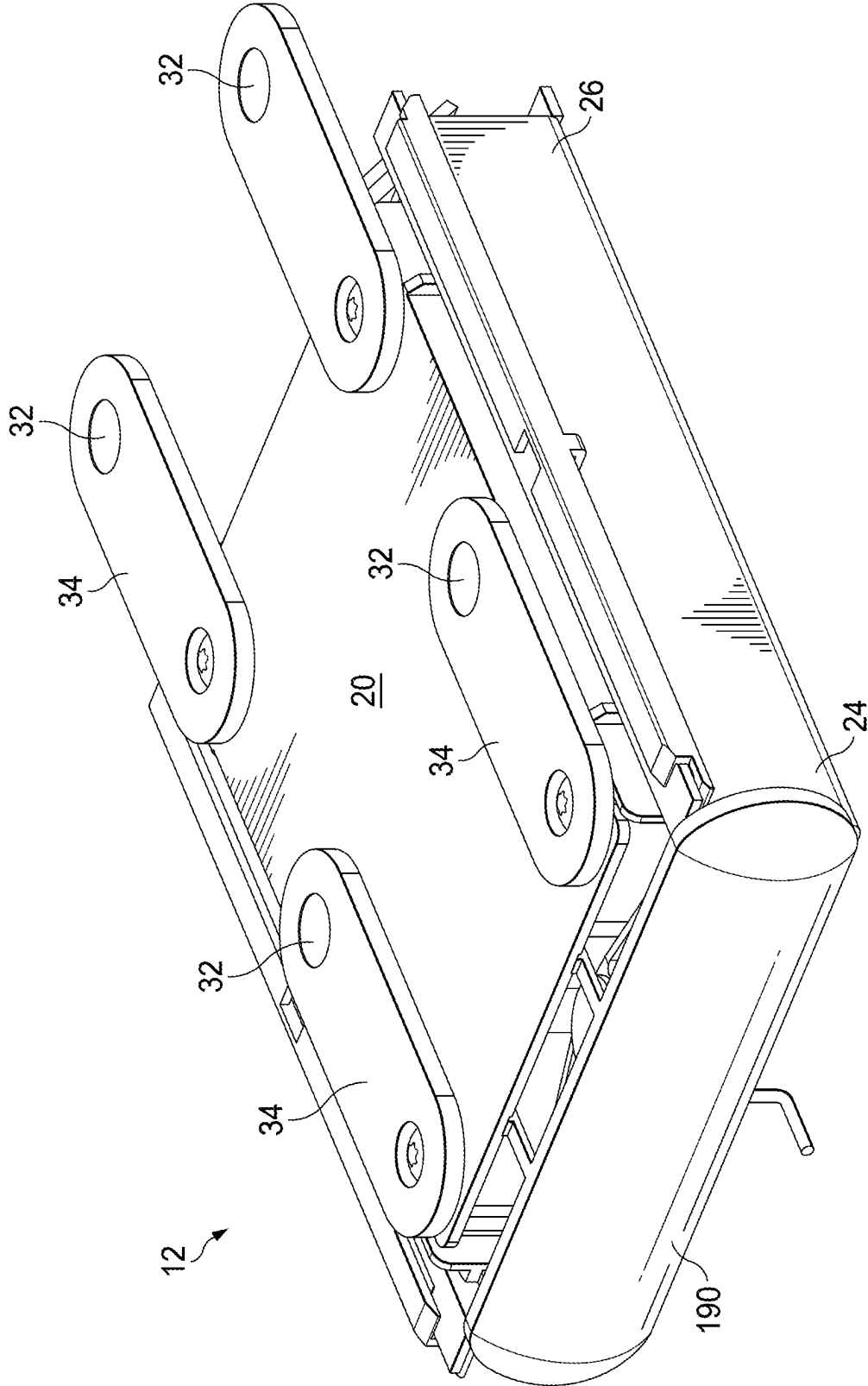


FIG. 15

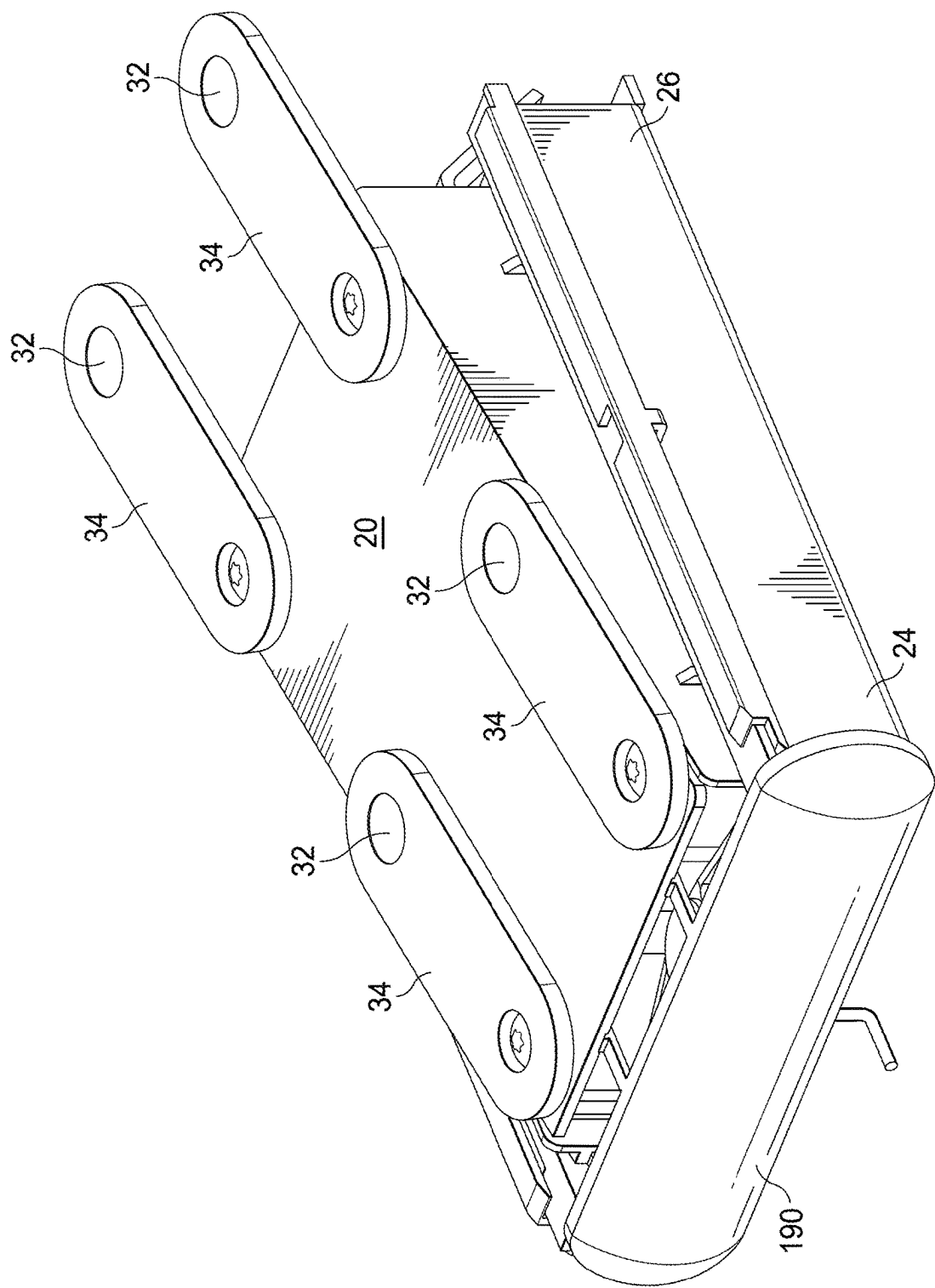
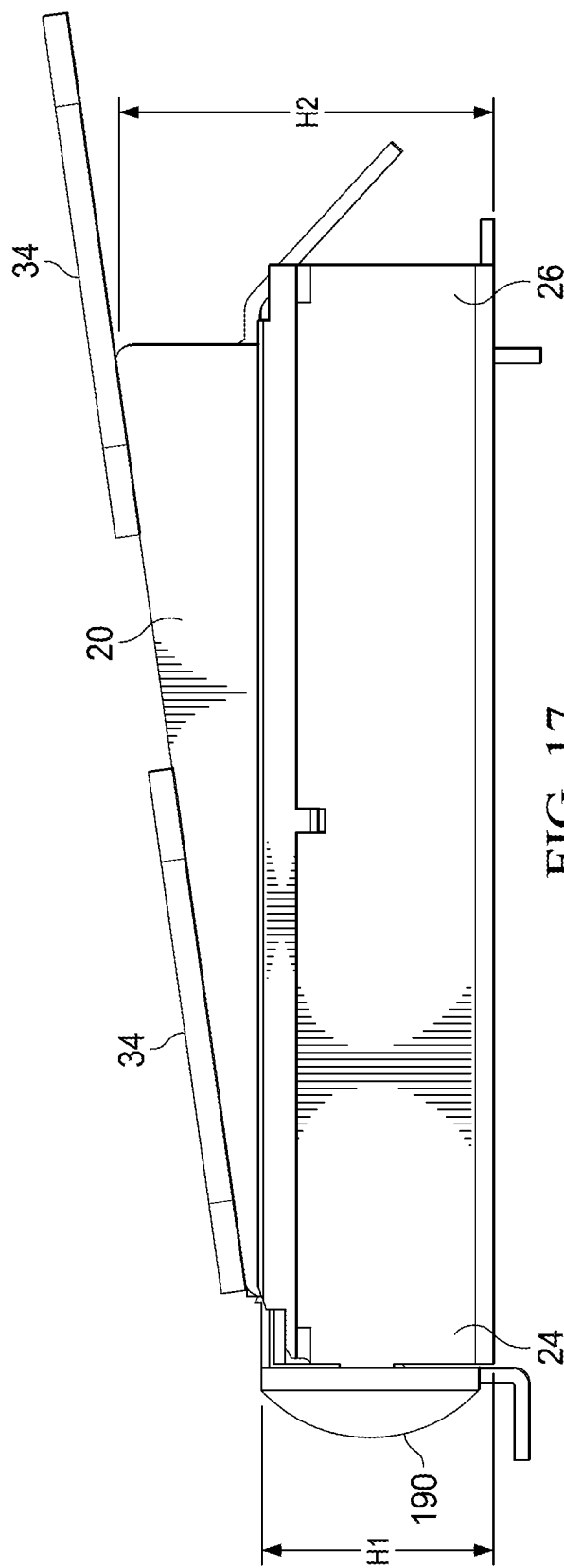
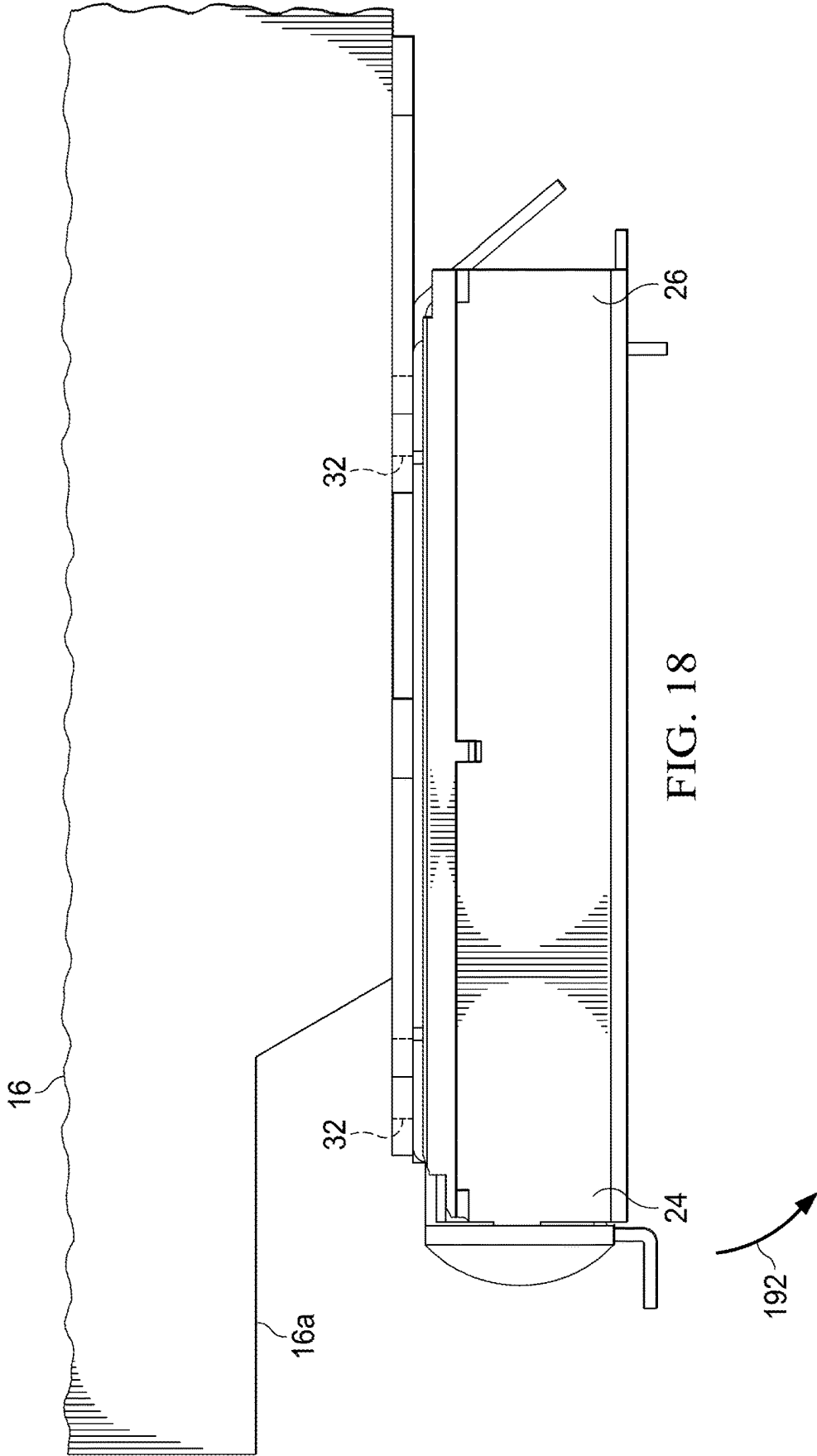


FIG. 16





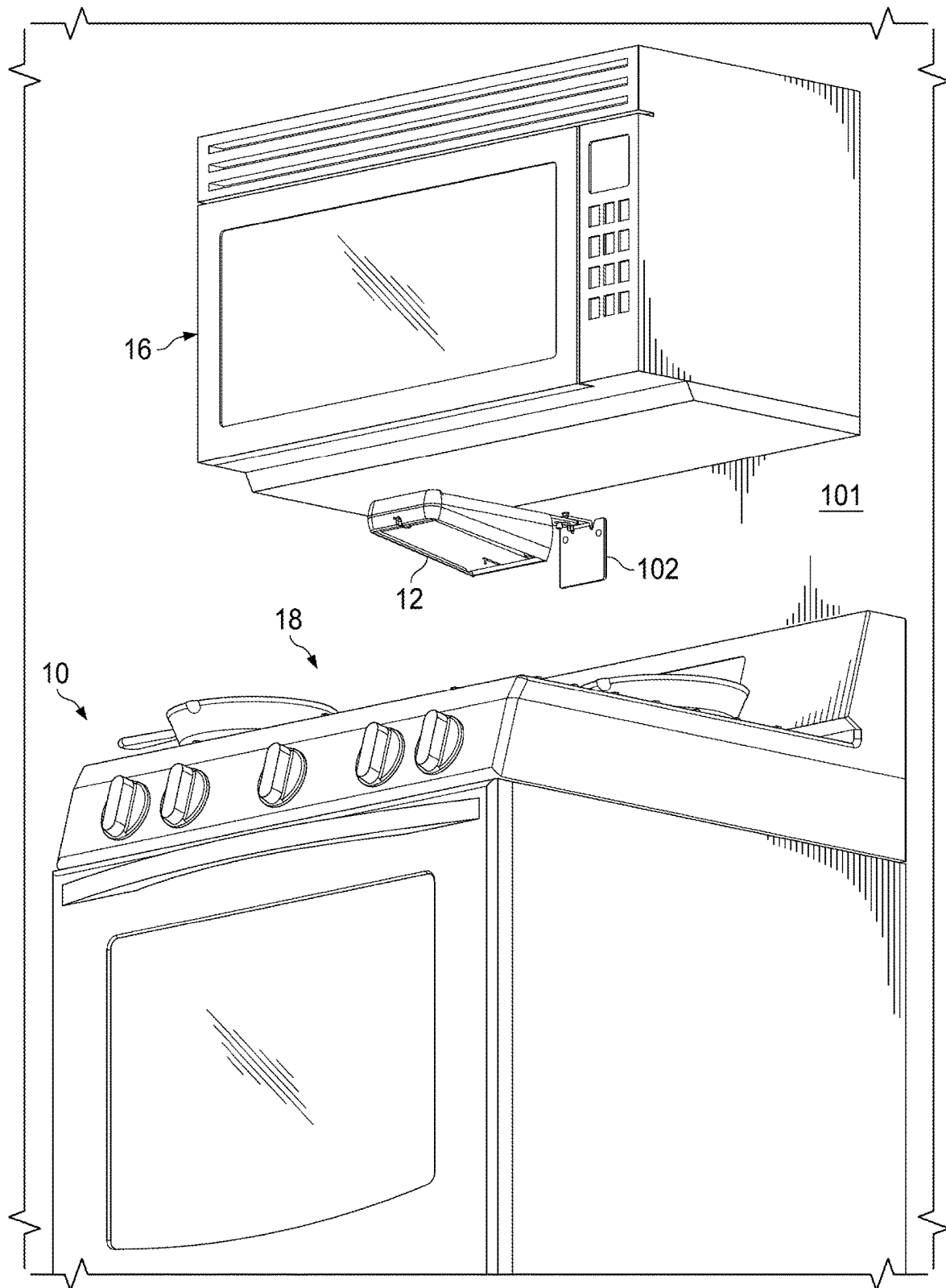


FIG. 19

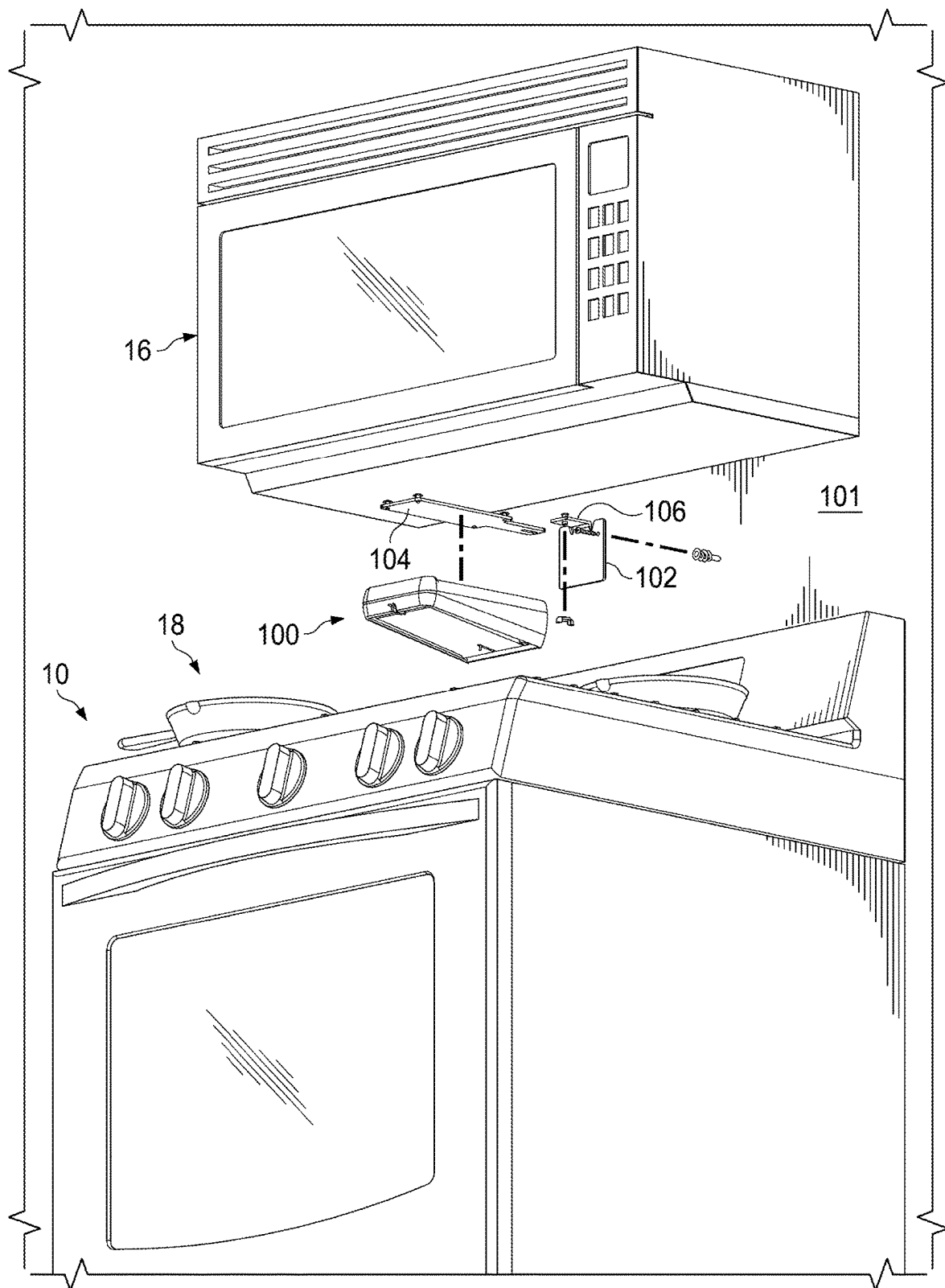


FIG. 20

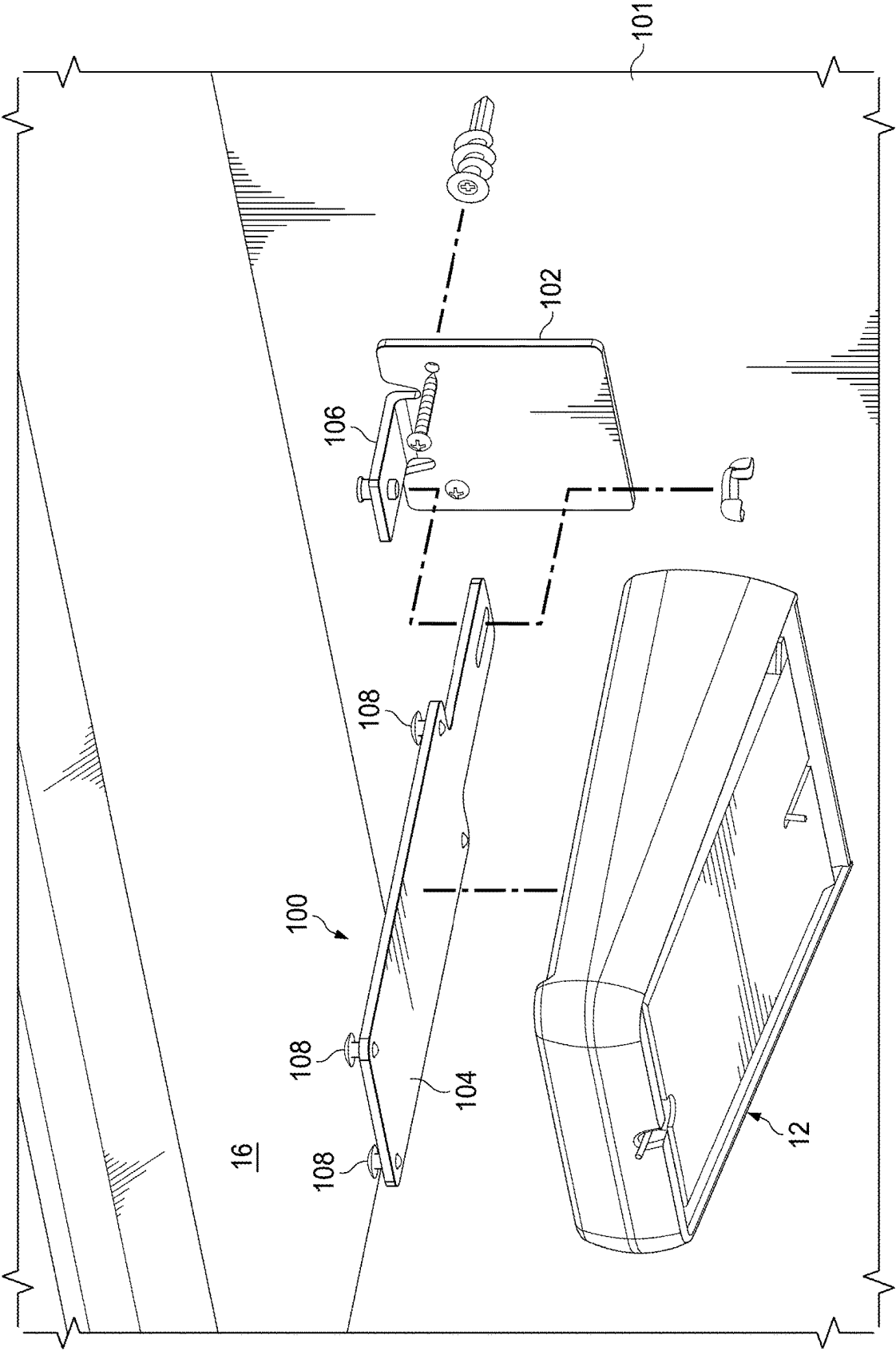


FIG. 21

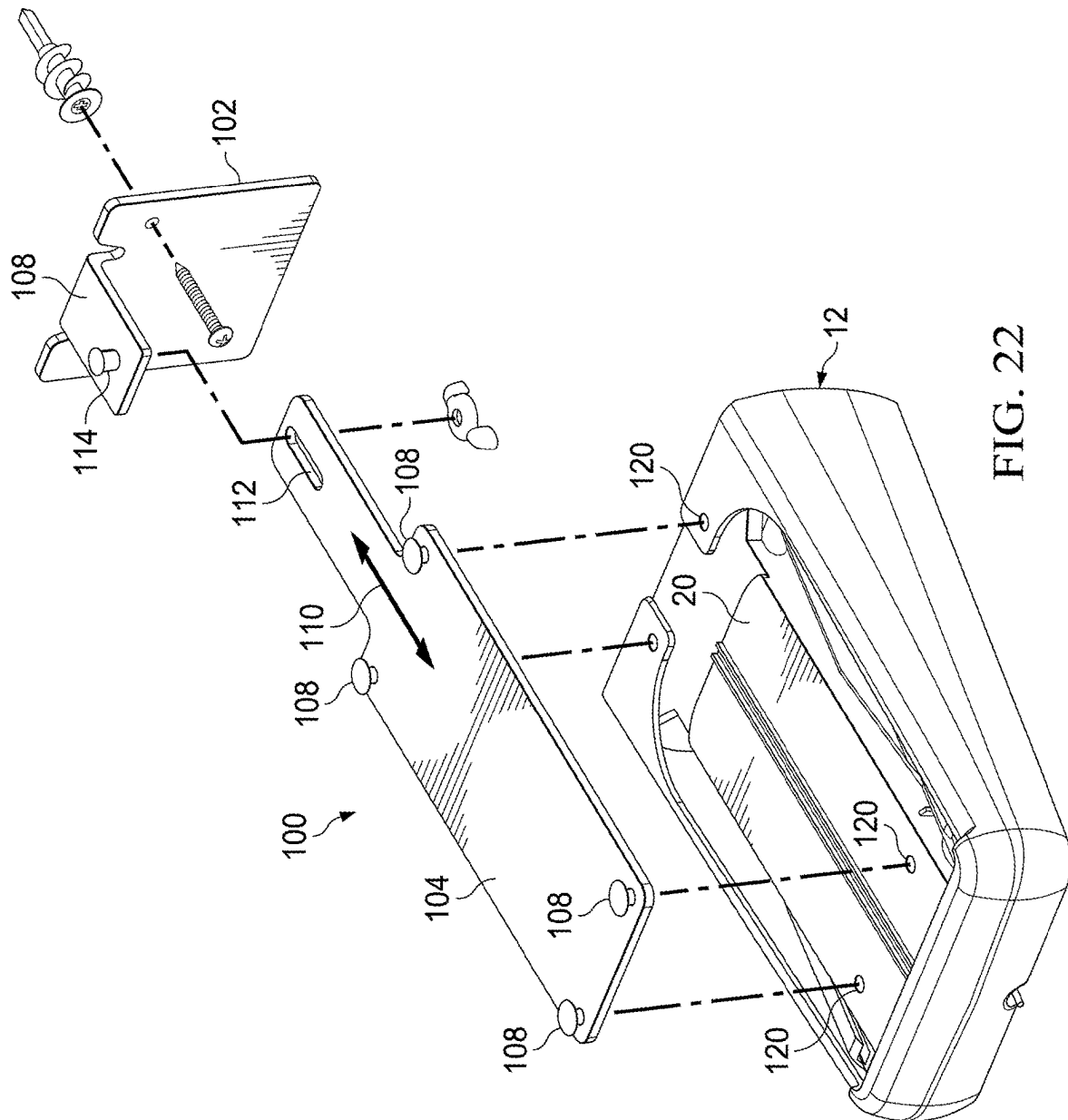


FIG. 22

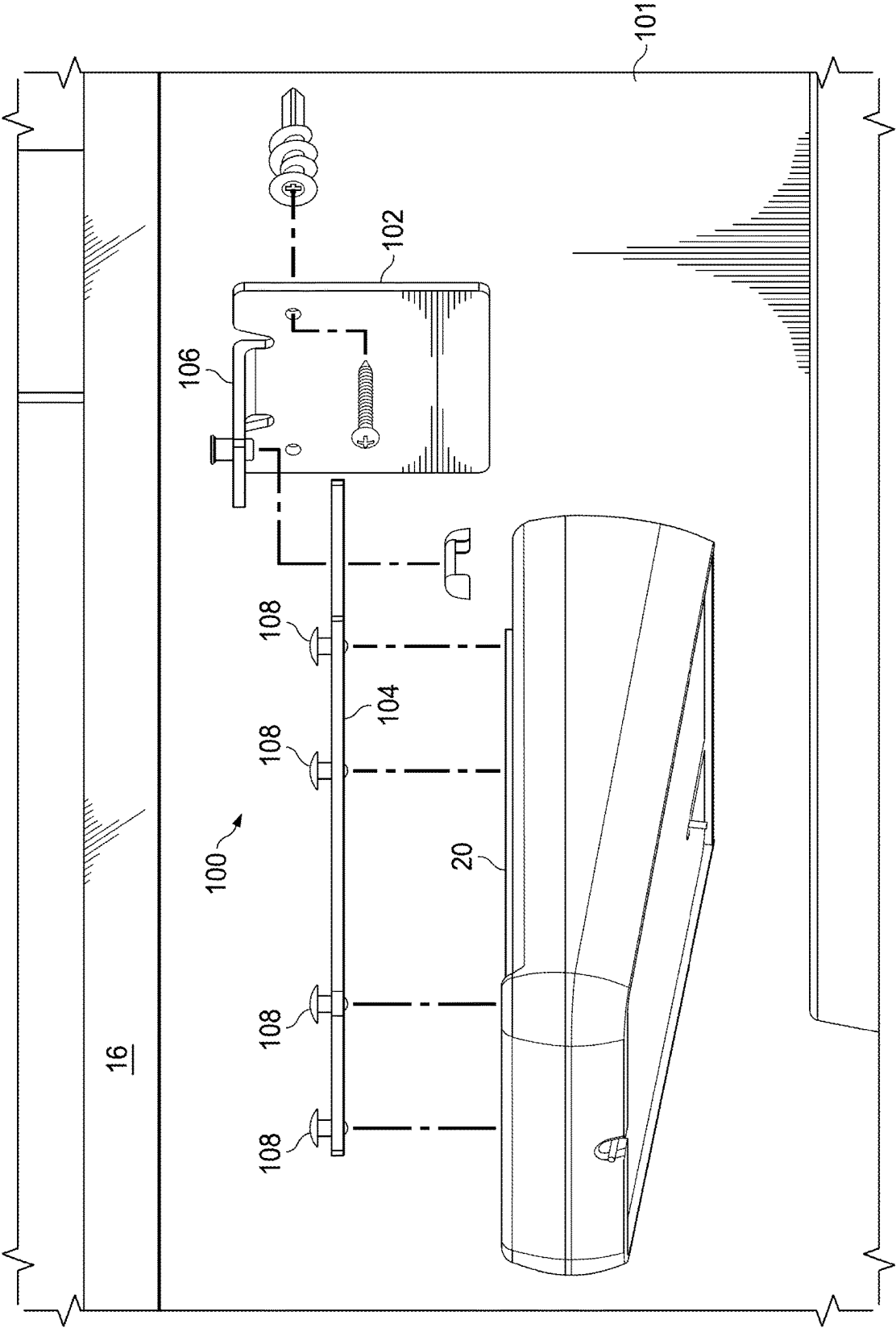


FIG. 23

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STOVETOP FIRE SUPPRESSION SYSTEM AND METHOD

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of and priority to U.S. Prov. Pat. App. No. 62/992,716 filed on Mar. 20, 2020, and entitled "Stovetop Fire Suppression System and Method," and U.S. Prov. Pat. App. No. 63/062,920 filed on Aug. 7, 2020, and entitled "Stovetop Fire Suppression System and Method," the disclosures of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

The present disclosure relates to a system and method for suppressing and extinguishing fires, and in particular, to an apparatus and method for mountable to a wall adjacent or near a stovetop cooking surface for suppressing and extinguishing kitchen stovetop fires.

BACKGROUND

In a kitchen, dangers are posed by fires originating from overheating of grease or oils during cooking. For example, a pan of grease on a stove burner can become so hot that it ignites. Not only does it pose a risk to individuals working in the area around the stove, but oftentimes, if not extinguished quickly, damage to the stove and the surrounding building structure can occur. This is especially true during circumstances where the stove is unattended. In this situation, considerable damage and devastation can occur, including the surrounding building structure becoming engulfed in flames.

In order to reduce the risk of household fires, fire suppression devices have been devised to automatically release a powder to suppress a stovetop fire, at least in part by smothering flames. Such a device is effective to put out a stovetop flame. Many designs have been proposed and used for extinguishing stovetop fires. However, such solutions are oftentimes difficult to install, are large, bulky, obstruct or otherwise block access to cooking surfaces, and are unable to store and/or precisely direct a sufficient amount of fire suppressant powder to extinguish a stovetop fire. It would be advantageous in the marketplace to address these and other deficiencies associated with current stovetop fire suppression devices

SUMMARY

According to a first aspect, there is provided stovetop fire suppression system for selectively releasing a fire suppressant powder onto a stovetop first or second burner. The stovetop fire suppression system includes a base member and a chute member secured to the base member at opposed first and second ends of the base member. The chute member is movable between a storage position, whereby the chute member and the base member form a storage area for storing the fire suppressant powder therein, and a dispensing position, whereby the chute member is positioned to release the fire suppressant powder from the storage area. The system further includes a first actuation member disposed at the base section first end and a second actuation member disposed at the base section second end. In response to actuation of the first actuation member, the chute member is movable the dispensing position to release the fire suppress-

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sant powder onto the first burner, and in response to actuation of the second actuation member, the chute member is moveable to the dispensing position to release the fire suppressant powder onto the second burner.

According to second aspect, there is provided a stovetop fire suppression system for selectively releasing a fire suppressant powder onto a stovetop first or second burner. The system includes base member having a top wall, a front wall, a rear wall, and a pair of sidewalls extending between the front and rear walls. The system also includes a chute member secured to the base member, the base member and the chute member forming a storage area for storing the fire suppressant powder. A first hinge is rotatably secured to the base member adjacent the base member front wall and a second hinge rotatably secured to the base member adjacent the base member rear wall. The chute member is secured to the first hinge and the second hinge at each respective end of the chute member; The system further includes a first actuation member disposed adjacent the base section front wall and a second actuation member disposed adjacent the base section rear wall such that in response to actuation of the first actuation member, the chute member moves to the dispensing position to release the fire suppressant powder onto the first burner, and in response to actuation of the second actuation member, the chute member moves to the dispensing position to release the fire suppressant powder onto the second burner.

According to third aspect, there is provided a method for suppressing a stovetop fire. The method includes providing a stovetop fire suppression system, the fire suppression system including a fire suppressant powder disposed within a storage area. The method further includes positioning the stovetop fire suppression system above a first burner and a second burner and securing stovetop fire suppression system to a support surface above the first burner and the second burner. The stovetop fire suppression system includes a first actuation member on a first end of the storage area and a second actuation member on an opposed second end of the storage area, the fire suppression system selectively releases the fire suppressant powder onto the first burner or the second burner in response to actuation of the first actuation member or the second actuation member, respectively.

DESCRIPTION OF THE FIGURES

The accompanying drawings facilitate an understanding of the various embodiments.

FIG. 1 is an illustration of a stovetop in which a stovetop fire suppression system is employed to advantage.

FIG. 2 is a front perspective view of the of stovetop fire suppression system FIG. 1 in a closed position.

FIG. 3 is a rear perspective view of the stovetop fire suppression system of FIGS. 1 and 2 in a dispensing position.

FIG. 4 is a front perspective view of the stovetop fire suppression system of FIGS. 1-3 in the dispensing position illustrating an extension member.

FIGS. 4a and 4b are illustrations of portion of the track and stop system for the extension member.

FIG. 5 is a rear perspective view of the stovetop fire suppression system of FIGS. 1-4 illustrating a fire suppressant powder stored in a storage area.

FIG. 6 is a bottom interior view of a base member of the fire suppression system of FIGS. 1-5.

FIG. 7 is a section view of the fire suppression system taken along the line 7-7 of FIG. 5.

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FIGS. 8-11 are illustrations of a section view of the fire suppression system of FIGS. 1-7 in the storage and dispensing positions.

FIGS. 12-14a are illustrations of the rotation stop and retaining system of the fire suppression system of FIGS. 1-11.

FIG. 15 is an illustration of the fire suppression system of FIGS. 1-14 having an end cover.

FIG. 16 is an illustration of another embodiment of a fire suppression system.

FIG. 17 illustrates a side view of the fire suppression system of FIG. 16.

FIG. 18 illustrates a fire suppression system secured to the underside of a microwave.

FIGS. 19 and 20 illustrate a fire suppression system secured to a wall via a wall support system.

FIG. 21 illustrates a bottom perspective view of the fire suppression system and the wall support system of FIGS. 19 and 20.

FIG. 22 illustrates top perspective view of the fire suppression system and the wall support system of FIGS. 19-21.

FIG. 23 illustrates an exploded view of the fire suppression system and the wall support system of FIGS. 19-22.

DETAILED DESCRIPTION

FIG. 1 is an illustration of a stovetop 10 in which a pair of fire suppression devices 12 are secured to the bottom side of an overhead support structure 16, such as, for example, a microwave or vent hood. As discussed in greater detail below, each fire suppression device 12 incorporates and/or otherwise employs a dual pivot/hinge system to enable each of the devices 12 to selectively open toward a front burner 15 or a rear burner 17 on a stovetop cooking surface 18 in response to detecting a fire on the front burner 15 or the rear burner 17.

Referring specifically to FIGS. 2 and 3, an exemplary embodiment of the fire suppression device 12 is configured in a storage position (FIG. 2) and in a dispensing position (FIG. 3). In FIGS. 2 and 3, the fire suppression device 12 includes a base member 20 and a chute member 22 rotatably secured to the base member 20 at first and second ends 24 and 26. In particular, the chute member 22 is rotatably secured to the base member 20 at the first end 24 so as to be movable between the storage position, whereby the fire suppressant powder 36 (FIG. 5) is stored within the suppression device 12, and the dispensing position, whereby the chute member 22 pivots downward toward the stovetop cooking surface 18 to dispense the fire suppression powder 36 onto the stovetop cooking surface 18.

According to embodiments disclosed herein, the fire suppression device 12 is releasably mounted to the underside of the overhead support structure 16 in a position such that (i) as the chute member 22 pivots about first end 24, the chute member 22 is aligned to dispense the fire suppression powder 36 directly onto the rear burner 17 and (ii) as the chute member 22 pivots about the second end 26, the chute member 22 is aligned to dispense the fire suppression powder 36 onto the front burner 15.

In the embodiment illustrated in FIGS. 1-4, the fire suppression device 12 is removably attached to the overhead support structure 16 via a plurality of magnets 32. The fire suppression device 12 includes a plurality of rotatable arms 34, each arm supporting at least one magnet 32 to facilitate attachment to the overhead support structure 16. Referring specifically to FIG. 2-4, the rotatable arms 34 are

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secured to the base member 20 via screws 37, which allows the rotational movement of the arm 34 in order to appropriately position the magnets 32 for attachment to the underside of the overhead support structure 16. The rotational movement provides flexibility to avoid non-metallic obstructions, such as, for example, lights and filters, that are often found on the underside of microwave ovens and vent hoods. It should be understood that other methods of attachment may be used, such as, for example, clips, in order to rotationally or non-rotationally secure the arms 34 to the base 20. The magnets 32 are strong enough to securely attach the fire suppression device 12 to the overhead support structure 16 without the need for additional or complicated mounting arrangements.

According to some embodiments, the clips can be integrally molded or otherwise a single unitary piece with the base member 20 so as to enable a secure connection between the rotatable arms 34 and the base member 20. According to some embodiments, the clips 37 comprise a plurality of spaced apart tabs that are sized to frictionally engage an inner wall of a corresponding opening 33 on the arm member 34. It should be understood, however, that any other type of method of securing the rotatable arms 34 to the base member 20 or any other portion of the fire suppression device 12 is permissible. Furthermore, while four rotatable arms 34 are illustrated, it should be understood that a greater or fewer number of rotatable arms 34 may be utilized and a greater number of magnets 32 may be secured to each rotatable arm 34. According to other embodiments, in addition to or in lieu of having rotatable arms 34, the arms can be designed with a slot to permit sliding movement of the arm 34 relative to the base member 20. Thus, for example, the arm 34 may have a slot sized to receive a corresponding pin extending from the base member 20, the pin engaging the slot to permit sliding movement of the arm 34 relative to the base member 20. Additionally, the arms 34, in addition to or in lieu of being rotatable, may be configured as adjustable in length to accommodate the obstructions commonly occurring on the underside of microwaves and vent hoods and other overhead support structures 16.

Referring specifically to FIGS. 3 and 4, the chute member 22 is formed having a bottom wall 50, a pair of sidewalls 50a and 50b and a metering bar 51 extending between the sidewalls 50a and 50b. According to some embodiments, the chute member 22 further includes an extension member 23 extendable from the chute member 22 such that an end 23a of the extension member 23 is positioned to direct fire suppression powder 36 onto the front burner 15. In the embodiment illustrated in FIGS. 3 and 4, the extension member 23 telescopically extends from the chute member 22. Similar to the chute member 22, the extension member 23 is formed having a bottom wall 90 and a pair of sidewalls 90a and 90b forming an extended ramp or pathway from the chute member 22 to the end 23a of the extension member 23. According to some embodiments, the extension member 23 includes a cross support member 90c extending between the sidewalls 90a and 90b. Cross support member 90c provides additional rigidity to the extension member 23.

In operation, as the chute member 22 transitions from the stored position downward toward the stovetop cooking surface 18 to the dispensing position, the extension member 23 telescopically extends to an extended position, as seen for example, in FIG. 4. Referring to FIGS. 3 and 4, the extension member 23 moves relative to the chute member 22 along tracks 80. Each track 80 is formed having a tongue 82 slideably disposed inside a groove 84. In the embodiment illustrated in FIG. 4, a tongue 82 extends outward from the

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sidewall **50a** and **50b**. The extension member **23** includes the groove **84** disposed along the sidewalls **90a** and **90b** sized to receive the respective tongue **82** therein. As seen specifically in FIGS. **4a** and **4b**, the groove **84** of the chute member **22** includes a stop system **99** formed as tab member **100** disposed at the first end **24**, the tab member **100** configured to contact and otherwise engage a corresponding end wall **85** of the groove **84** on the extension member **23** so as to prevent overextension of the extension member **23** in the direction of arrow **106**. In the embodiment illustrated in FIGS. **4a** and **4b**, the chute member **22** includes a tab member **100** on each tongue **82** extending from sidewalls **50a** and **50b**; however, it should be understood that only a single tab member **100** on either sidewall **50a** or **50b** may be utilized. Furthermore, it should be understood the stop system **99** may consist of any other configuration designed or otherwise utilized to resist the unwanted movement of the extension member **23** when the fire suppression device **12** actuates to extinguish a fire on a rear burner **17**.

It should be understood that other methods of extension may be used. For example, the extension member **23** may be pivotably attached to the chute member **22**.

Referring to FIGS. **3**, **5** and **6**, the fire suppression powder **36** is illustrated stored within a storage area **28** that is sized to store a sufficient volume of fire suppression powder **36** to suppress at least one stovetop fire. As illustrated, the base member **20** and the chute member **22**, when the fire suppression device is in the closed position, form the storage area **28** for holding the fire suppression powder **36** therein. In particular, the base member **20** includes a top wall **40** (removed from FIG. **5**, but illustrated in FIGS. **2-4**), a pair of sidewalls **42** and **44**, a front wall **46** and a rear wall **48** extending from the top wall **40** so as to contact and otherwise engage the bottom wall **50** (best illustrated in FIG. **3**) of the chute member **22**. With particular reference to FIG. **3**, the chute bottom wall **50** includes a channel **52** to receive a seal **54** (see FIG. **8**). The channel **52**, and thus the seal **54**, correspond to the shape of the walls **42**, **44**, **46** and **48** of the base member **20** such that when the fire suppression device **12** is in the closed position, the walls **42**, **44**, **46** and **48** sealingly engage the seal **54** in order to contain the fire suppression powder **36** inside the storage area **28**.

Referring now to FIGS. **5-7**, a pair of hinges **60** and **62** are disposed at respective first and second ends **24** and **26** to not only enable movement of the chute member **22** between the stored and discharge positions, but also, as explained in greater detail below, to facilitate the direction of discharge. In FIG. **6**, hinges **60** and **62** extend between sidewalls **42** and **44** of the base member **20**. In some embodiments, each end of the hinge **60** and **62** includes a boss **110** supported within an opening **112** on each sidewall **42** and **44** to enable the rotational movement of the chute member **22**. For example, hinges **60** and **62** are positioned at respective first and second ends **24** and **26** such that in response to an actuation device **120a** detecting a fire on the front burner **15**, the chute rotates about hinge **62** to direct fire suppression powder **36** on the front burner **15**. Likewise, in response an actuation device **120b** detecting a fire on the rear burner **17**, the chute rotates about hinge **60** to direct fire suppression powder **36** onto the rear burner **17**.

Referring now to FIGS. **5-8**, the hinges **60** and **62** support respective actuation devices **120a** and **120b**. Each hinge **60** and **62** retains and otherwise supports the actuation device **120a** and **120b**, which secures the chute member **22** to the base member **20** in the closed position. In operation, actuation devices **120a** and **120b** are used to not only detect a fire condition, but when such condition is detected, the actuation

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devices **120a** and **120b** position the fire suppression device **12** from the closed position to the discharge position by causing the separation of, as explained in greater detail below, the chute member **22** from the base member **20**.

In the embodiment illustrated in FIGS. **5-8**, each actuation device **120a** and **120b** is formed having a charge cup **122** secured to each hinge **60** and **62** within an actuator support opening **124** (best illustrated in FIG. **6**) and a corresponding charge cap **126** is attached to the each end of the chute member **22** on the bottom wall **50**. The charge cap **126** is designed to be releasably secured to the charge cup **122** to facilitate separation therebetween. According to embodiments disclosed herein, the charge cap **126** threadingly engages the charge cup **122** in order to secure the chute member **22** to the base member **20**. It should be understood, however, that other means of securing the charge cap **126** to the charge cup **122** can be used. For example, the charge cap **126** can be frictionally secured to the charge cup **122**, an adhesive can be used to secure the charge cap **126** to the charge cup **122**, any other method that allows releasable attachment between the parts.

As illustrated in FIGS. **7** and **8**, the charge cap **126** and the charge cup **122** form an interior cavity **128** for storing gun powder therein (not illustrated). A fuse **130** extends from the interior cavity **128** outside to the surrounding area such that when the fire suppression device **12** is disposed over the stovetop cooking surface **18**. If a fire were to occur, the fuse **130** would burn, ultimately igniting the gun powder in the interior cavity **128**, causing the chute member **22** to separate from the base member **20**.

Referring now to FIGS. **8** and **9**, when a fire occurs on the rear burner **17** (FIG. **1**), the flames light the fuse **130** over the rear burner **17**, which ignites the gunpowder disposed in the interior cavity **128**. The ignition of the gun powder creates a separation force **F1** resulting in separation of the chute member **22** from the base member **20** at the second end **26** causing rotation of the chute member **22** about the hinge **60** in the direction of arrow **132**. The front actuation device **120a** remains intact and is retained and otherwise supported by the hinge **60** allowing the chute member **22** to rotate in the direction of arrow **132**. As the chute member **22** is rotated, the fire suppression powder **36** is released and gravitally flows from the storage area **28** and guided toward the fire, and in particular, the rear burner **17**, by the chute member **22**. Flow is restricted by an opening **53** formed by the metering bar **51**, which causes the fire suppression powder **36** to fall onto the fire at a controlled rate.

Likewise, when a fire occurs on the front burner **15**, flames light the fuse **130** on the first end **24**, which ignites the gunpowder in the interior cavity **128**. As illustrated in FIGS. **10** and **11**, the ignition of the gun powder creates a separation force **F2** resulting in separation of the chute member **22** from the base member **20** at the first end **24**, causing rotation of the chute member **22** about the rear hinge **62** in the direction of arrow **134**. As the chute member **22** is rotated away from the base member **20**, the extension member **23** extends outward in the direction of arrow **55** so as to direct the fire suppression powder **36** over the front burner **15**.

A chute push bar **180** extends from the base member **20** and is sized to contact a chute extension **182** that extends from the chute member **22**. In operation, when the actuation member **120a** actuates causing the chute member **22** to rotate in the direction of arrow **134**, the extension **23** extends outward in response to the chute extension **182** contacting the chute push bar **180**. In other embodiments. However, other methods of extension may be used. For example, in

other embodiments, a biasing member (not illustrated), such as a spring, may be utilized to exert a force on the extension member 23 to force it in the direction of arrow 55 to the extended position. Thus, in response to the detection of a fire, as the chute member 22 transitions to the dispensing position, the stored energy in the biasing member exerts a force on the extension member 23 to position the extension member 23 in the extended position. In addition or in lieu of the foregoing, the chute member 23 is operable move to the extended position via gravity acting on the chute member 23.

Referring now to FIGS. 12-14, with detailed views shown as FIG. 12a, FIG. 12b, FIG. 13a, and FIG. 14a, the first and second ends 24 and 26 each optionally include a rotation stop 150 and a retention system 152 for preventing over rotation of and separation of the chute member 22 from the fire suppression system 12. In the embodiment illustrated in FIGS. 12-14 rotation stops 150 are formed or otherwise disposed on the base sidewalls 42 and 44 to limit the rotation of the chute member 22. In addition, base upper guides 162 and base lower guides 164 are also formed on or otherwise disposed on the sidewalls 42 and 44 of the base 20. Corresponding retaining guides 166 are located on the chute member sidewalls 50a and 50b and are positioned between and slideably engage the base upper and lower guides 162 and 164 on the base member 20. In use, the base upper and lower guides 162 and 164, in addition to providing a guide for rotation of the chute member 22, the base upper and lower guides 162 and 164 prevent separation of the chute member 22 from the base member 20 in the event that both actuation devices 120a and 120b actuate.

FIGS. 13 and 14 illustrate the rotation of the chute member 22. In FIG. 13, as the chute member 22 rotates so as to extinguish a fire on the rear burner 17, the base upper and lower guides 162 and 164 disposed at the first end 24 engage the corresponding retaining guides 166. The chute member 22 rotates in the direction of arrow 132 until the retaining guides 166 contact rotation stops 150, at which point rotation stops. This also prevents separation of the chute member 22 from the base member 20.

Referring to FIG. 14, as the chute member 22 rotates to the front open position to dispense the fire suppression powder 36 on the front burner, the base upper and lower guides 162 and 164 disposed on the second end 26 engage the corresponding retaining guides 166 on the base member 20. The chute member 22 rotates in the direction of arrow 134 until the retaining guides 166 contact the rear rotation stops 150. This prevents further rotation and prevents separation of the chute member 22 from the base member 20.

Turning to FIG. 15, an embodiment of the fire suppression system 12 is illustrated having an end cap or cover 190 position to cover the first end 24. According to embodiments disclosed herein, the cover 190 is secured to the fire suppression system 12 via a snap/friction fit, an adhesive or any other method, such as screws, that can securely fasten the cover 190 to the fire suppression system 12. According to some embodiments, the cover 190 can be sized to extend around the sides of the fire suppression system 12, can have a height greater than the height of the fire suppression system 12, and further, a single cover can extend between multiple fire suppression systems 12 when mounted to an overhead support surface 16. The cover 190 is optional and functions to provide an aesthetic appearance to the suppression system 12.

FIGS. 16 and 17 illustrate another embodiment of the fire suppression system 12 in which the base member 20 is formed having a varying height. For example, the base

member 20 varies in height from the first end 24 to the second end 26. In FIGS. 16 and 17, the height of the base 20 member gradually increases from the first end 24 to the second end 26 forming a wedge shape appearance. Significant advantages result from this wedge shape. For example, the width W of the fire suppression system 12 can be narrowed in this embodiment (as compared to the embodiment illustrated in FIGS. 1-15) so as to store the same volume of powder 36 inside the storage area 28. In addition, because an increased volume of powder 36 is stored closer to the second end 26 rather than evenly distributed between the first and second ends 24 and 26, the center of gravity of the fire suppression system 12 is closer to the second end 26. This is important in situations where the magnets 32 closer to the first end 24 are unable to attach to an overhead support structure 16. For example, referring to FIG. 18, oftentimes the forward undersides of a microwave includes a recessed portion 16a to enable unobstructed viewing of the cooking surface 18. As a result, a suitable surface is not available for engagement with the magnets 32 located toward the first end 24. Because the center of gravity is rearward and closer to the second end 26, the magnets 32 closer to the second end 26 are able to support the fire suppression system 12 and resist the moment in the direction of arrow 192 since the moment is reduced due to a lesser amount of powder, and thus weight, cantilevering away from the support structure 16. Furthermore, by having a narrower profile due to the wedge shape, to the extent the fire suppression system 12 covers a vent or light on the bottom of the microwave, the narrower profile will reduce the total overlap. In addition to the above, by having the center of gravity closer to the second end, as the system 12 transitions to the dispensing position, the release of powder 36 occurs in a more controlled fashion, especially as the chute 22 and extension 23 are fully extended.

It should be understood that the size of the fire suppression system 12 may vary. For example, the length L is approximately 6 inches and the width W is approximately 5 inches. However, it should be understood that these lengths may be different depending on the size of the stovetop cooking surface 18, including the position of burners 15 and 17, and the size of the overhead support surface 16. For example, according to some embodiments, the length L may be selected between and including a range between 5 inches and 12 inches. And the width W may be selected between and including a range between 2 inches and 6 inches. It should be understood that in other embodiments, the length L and width W may be outside of these ranges. Furthermore, the height H1 is about 1.5 inches, however, it should be understood that height H1 may be otherwise (i.e., taller or shorter). Referring to FIGS. 17 and 18, in some embodiments, the height of the wedge shaped base 20 at the first end 24 is 1.5 inches and the height H2 at the second end is 2.25 inches, although the heights at both ends 24 and 26 may be different (i.e., taller or shorter).

Referring now to FIGS. 19-23, a stovetop support system 100 is employed to secure the fire suppression device 12 to an adjacently position wall or other support surface 101, rather than the underside of the microwave 16, such that the suppression device is disposed above the stovetop 18. Referring specifically to FIGS. 21 and 22, the wall support system 100 includes a base member 102 to be secured to the wall and a bracket member 104 configured to extend and otherwise cantilever outward from the base member 104. In the embodiment illustrated in FIGS. 21 and 22, the bracket member 104 is removably attachable to the base member 102 via an arm 106 that cantilevers outward from the base

member **102**. Referring specifically to FIG. **22**, the connector arm **106** is formed integrally with the base member **102** and the bracket member **104** is removably detachable to the arm **106**. However, it should be understood that the wall support system **100** may be otherwise formed. For example, the wall support system **100** may be a single unitary piece attachable to the wall, or the bracket member **104** can be integrally formed with the connector arm **106**.

Referring specifically to FIG. **22**, the fire suppression device **12** is configured to be secured to the bracket member **104** via a plurality of screws **108**. In the embodiment illustrated in FIGS. **22** and **23**, four screws **108** are used to secure the fire suppression device **12** to the bracket **104**; however, it should be understood that a greater or fewer number of screws **108** can be used. During installation openings **120** on the fire suppression device are aligned with corresponding openings on the bracket **104** so as to secure the fire suppression device **12** to the bracket **104** via the screws **108**. It should be further understood that other attachment mechanisms may be used. For example, one or more magnets (not illustrated) could be used to secure the fire suppression device **12** to the wall support system **100**. In other embodiments, an adhesive may be used.

With continued reference to FIG. **22**, the position of the bracket **104** may be adjusted toward or away from the wall in the direction of arrows **110** via a slot **112**. In the embodiment illustrated in FIG. **22**, the slot **112** is formed on the bracket **104**; however, the slot **112** may be formed on the arm **106** or on both the bracket **104** and the arm **106**. Regardless of the configuration, a screw **114** can be used to adjust and secure the bracket **104** relative to the arm **106** so as to adjust the position of the fire suppression device **12** over the stovetop **18** (FIG. **1**).

According to other embodiments, the wall support system **100** may be integrally formed with the fire suppression device **12**. For example, the wall support system **100** may extend from a rear wall **118** of the first suppression device **12** to facilitate securing the fire suppression device to the wall.

Embodiments disclosed in FIGS. **19-23** enable the fire suppression device to be mounted above the stovetop cooking surface **18** without interfering with vents, lighting and having to accommodate irregular mounting surfaces often-times occur with overhead support structures **16** such as microwaves and vent hoods.

In the foregoing description of certain embodiments, specific terminology has been resorted to for the sake of clarity. However, the disclosure is not intended to be limited to the specific terms so selected, and it is to be understood that each specific term includes other technical equivalents which operate in a similar manner to accomplish a similar technical purpose.

In the specification and claims, the word “comprising” is to be understood in its “open” sense, that is, in the sense of “including”, and thus not limited to its “closed” sense, that is the sense of “consisting only of”. A corresponding meaning is to be attributed to the corresponding words “comprise”, “comprised” and “comprises” where they appear.

In addition, the foregoing describes only some embodiments of the invention(s), and alterations, modifications, additions and/or changes can be made thereto without departing from the scope and spirit of the disclosed embodiments, the embodiments being illustrative and not restrictive.

Furthermore, invention(s) have described in connection with what are presently considered to be the most practical and preferred embodiments, it is to be understood that the

invention is not to be limited to the disclosed embodiments, but on the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the invention(s), as defined solely by the appended claims. Also, the various embodiments described above may be implemented in conjunction with other embodiments, e.g., aspects of one embodiment may be combined with aspects of another embodiment to realize yet other embodiments. Further, each independent feature or component of any given assembly may constitute an additional embodiment.

What is claimed is:

1. A stovetop fire suppression system for selectively releasing a fire suppressant powder onto a stovetop first or second burner, the stovetop fire suppression system comprising:

a base member and a chute member secured to the base member at opposed first and second ends of the base member, the chute member movable between a storage position, whereby the chute member and the base member form a storage area for storing the fire suppressant powder therein, and a dispensing position, whereby the chute member is positioned to release the fire suppressant powder from the storage area;

a first heat sensitive member disposed at the base section first end;

a second heat sensitive member disposed at the base section second end; and

wherein in response to actuation of the first heat sensitive member being exposed to a pre-determined temperature, the chute member is movable to the dispensing position to release the fire suppressant powder onto the first burner, and in response to actuation of the second heat sensitive member being exposed to the pre-determined temperature, the chute member is movable to the dispensing position to release the fire suppressant powder onto the second burner.

2. The stovetop fire suppression system of claim 1, wherein the chute member is pivotably secured to the base member at a base member first end and a base member second end.

3. The stovetop fire suppression system of claim 1, further comprising an extension member extending from the chute member in response to the second heat sensitive member being exposed to the pre-determined temperature to dispense the fire suppressant powder onto the second burner.

4. The stovetop fire suppression system of claim 3, wherein the extension member telescopically extends from the chute member.

5. The stovetop fire suppression system of claim 1, further comprising a metering bar extending laterally across the chute member to control the flow of the fire suppressant powder onto the applicable first or second burner.

6. The stovetop fire suppression system of claim 1, further comprising a plurality of support arms secured to the base member for securing the fire suppression system to an overhead support structure disposed above the first and second burners.

7. The stovetop fire suppression system of claim 6, wherein the plurality of support arms are pivotably secured to the base member.

8. The stovetop fire suppression system of claim 6, wherein the overhead support structure is a microwave oven or a vent hood.

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9. The stovetop fire suppression system of claim 6, wherein the plurality of support arms further include a magnet for securing the suppression system to an overhead support surface.

10. A stovetop fire suppression system for selectively releasing a fire suppressant powder onto a stovetop first or second burner, the stovetop fire suppression system comprising:

a base member having a top wall, a front wall, a rear wall, and a pair of sidewalls extending between the front and rear walls;

a chute member secured to the base member, the base member and the chute member forming a storage area for storing the fire suppressant powder;

a first hinge rotatably secured to the base member adjacent the base member front wall and a second hinge rotatably secured to the base member adjacent the base member rear wall, the chute member secured to the first hinge and the second hinge at each respective end of the chute member;

a first heat sensitive member disposed adjacent the base section front wall and a second heat sensitive member disposed adjacent the base section rear wall; and

wherein in response to the first heat sensitive member being exposed to a pre-determined temperature, the chute member moves to the dispensing position to release the fire suppressant powder onto the first burner, and in response to the second heat sensitive member being exposed to the pre-determined temperature, the chute member moves to the dispensing position to release the fire suppressant powder onto the second burner.

11. The stovetop fire suppression system of claim 10, wherein the heat sensitive members include a fuse, a charge cup and a charge cap, the charge cup and the charge cap forming an interior area to receive a portion of the fuse, such that upon actuation of the actuation member, the charge cup separates from the charge cap.

12. The stovetop fire suppression system of claim 10, further comprising an extension member extending from the chute member in response to the second heat sensitive member being exposed to the pre-determined temperature to dispense the fire suppressant powder onto the second burner.

13. The stovetop fire suppression system of claim 12, wherein the extension member telescopically extends from the chute member.

14. The stovetop fire suppression system of claim 10, wherein the chute member further comprises a pair of

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sidewalls and at least one metering bar extending laterally across the chute member to control the flow of fire suppressant powder onto the applicable first burner or the second burner.

15. The stovetop fire suppression system of claim 10, further comprising at least one retaining arm pivotably secured to the base member for securing the fire suppression system to a support surface disposed above the first and second burners.

16. The stovetop fire suppression system of claim 15, wherein the at least one retaining arm further includes a magnet for securing the fire suppression system to the support surface.

17. The stovetop fire suppression system of claim 10, wherein a height of the base member at the second end is greater than the height of the base member at the first end.

18. The stovetop fire suppression system of claim 17, wherein the height of the base member at the second end is 2.25 inches.

19. A stovetop fire suppression system for selectively releasing a fire suppressant powder onto a stovetop burner, the stovetop fire suppression system comprising:

a base member having a first end and an opposed second end;

a chute member having a bottom wall and rotatably secured adjacent the base member first end, the chute member and the base member forming a storage area for storing the fire suppressant powder;

an extension member having a bottom wall and slidably engaged with the chute member;

a heat sensitive member disposed adjacent the base member second end; and

wherein in response to the heat sensitive member being exposed to a pre-determined temperature, the chute member rotates from a storage position to a dispensing position and the extension member slidably and telescopically extends from the chute member such that the bottom wall of the extension member and the bottom wall of the chute member are disposed adjacent to each other and form an extended and continuous ramp extending from the chute member to direct and release the fire suppressant powder onto the stovetop burner.

20. The stovetop fire suppression system of claim 19, wherein the chute member further comprises a pair of sidewalls and at least one metering bar extending laterally across the chute member to control the flow of fire suppressant powder.

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